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Silicone Wire Embolization-induced Acute Retinal Artery Ischemia and Reperfusion Model in Mouse: Gene Expression Provides Insight into Pathological Processes

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Abstract

Acute retinal ischemia and ischemia-reperfusion injury are the primary causes of retinal neural cell death and vision loss in retinal artery occlusion (RAO). The absence of an accurate mouse model for simulating the retinal ischemic process has hindered progress in developing neuroprotective agents for RAO. We developed a unilateral pterygopalatine ophthalmic artery occlusion (UPOAO) mouse model using silicone wire embolization combined with carotid artery ligation. The survival of retinal ganglion cells and visual function were evaluated to determine the duration of ischemia. Immunofluorescence staining, optical coherence tomography, and haematoxylin and eosin staining were utilized to assess changes in major neural cell classes and retinal structure degeneration at two reperfusion durations. Transcriptomics was employed to investigate alterations in the pathological process of UPOAO following ischemia and reperfusion, highlighting transcriptomic differences between UPOAO and other retinal ischemia-reperfusion models. The UPOAO model successfully replicated the acute interruption of retinal blood supply observed in RAO. 60-minutes of Ischemia led to significant loss of major retinal neural cells and visual function impairment. Notable thinning of the inner retinal layer, especially the ganglion cell layer, was evident post-UPOAO. Temporal transcriptome analysis revealed various pathophysiological processes related to immune cell migration, oxidative stress, and immune inflammation during the non-reperfusion and reperfusion periods. A pronounced increase in microglia within the retina and peripheral leukocytes accessing the retina was observed during reperfusion periods. Comparison of differentially expressed genes (DEGs) between the UPOAO and high intraocular pressure models revealed specific enrichments in lipid and steroid metabolism-related genes in the UPOAO model. The UPOAO model emerges as a novel tool for screening pathogenic genes and promoting further therapeutic research in RAO.

eLife assessment

The manuscript establishes a sophisticated mouse model for acute retinal artery occlusion (RAO) by combining unilateral pterygopalatine ophthalmic artery occlusion (UPOAO) with a silicone wire embolus and carotid artery ligation, generating ischemia-reperfusion injury upon removal of the embolus. This clinically relevant model is **useful** for studying the cellular and molecular mechanisms of RAO. The data overall are **solid**, presenting a novel tool for screening pathogenic genes and promoting further therapeutic research in RAO.

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1 Introduction

Retinal artery occlusion (RAO) is a severe ophthalmic disease characterized by a sudden interruption of blood flow in the retinal artery, leading to retinal ischemia [1]. Over 60% of RAO patients suffer from impaired vision, ranging from finger counting to complete vision loss [2]. Additionally, RAO patients face increased risk of cardiovascular and cerebrovascular events [3–5]. Retinal ischemia and hypoxia can lead to irreversible damage to retinal cells within 90 minutes [6]. This damage is primarily due to the apoptosis of retinal ganglion cells (RGCs), which is driven by inflammation and oxidative stress. Unfortunately, conservative treatments for RAO, such as ocular massage and hyperbaric oxygen therapy, provide limited therapeutic benefits [7]. Although thrombolysis is effective, its application is restricted by a narrow therapeutic window (typically within 4.5 hours) [8]. Furthermore, the restoration of blood flow to the ischemic area following thrombolysis can cause a sudden increase in tissue oxidative levels, resulting in ischemia-reperfusion injury (IRI) in the retina [9]. IRI is a common pathological condition that can induce inflammation, retinal tissue damage and visual impairment [10]. Therefore, it is crucial to develop an animal model that accurately simulates the pathological processes of RAO to thoroughly investigate the pathophysiological changes and explore potential neuroprotective treatments following retinal ischemia and reperfusion.

Various animal models have been employed to investigate the effects of retinal injury resulting from ischemia and subsequent reperfusion in RAO [6, 11, 12]. Based on the methods used for modelling, retinal ischemia-reperfusion models can be categorized into two groups: 1) intravascular occlusion models, including photochemical-induced thrombosis model and vascular intervention model; and 2) extravascular occlusion models, including central retinal artery ligation (CRAL) model, unilateral common carotid artery occlusion (UCCAO) model, and high intraocular pressure (HIOP) model. The photochemical-induced thrombosis model and vascular intervention model have been reported to be useful tools for assessing retinal injury after ischemia and reperfusion. However, both models are limited by the need for trained interventional radiologists and the need for advanced techniques [13, 14]. The CRAL model, which directly clips or ligates the central retinal artery, also presents drawbacks, such as potential optic nerve damage and difficulty in application to small animals such as mice, limiting its use in experiments with larger sample sizes [6]. The UCCAO model, which induces retinal hypoperfusion by occluding the unilateral common carotid artery (CCA), is more suitable for simulating chronic ischemic retinal disease than acute retinal ischemia [15, 16]. The HIOP model is widely utilized in mice for studying ischemia-reperfusion in acute primary angle-closure glaucoma (APACG) [17]. However, the mechanical compression of the retina caused by saline

injection in this model may also lead to retinal damage, potentially influencing IRI research. Therefore, it is critically important to develop a simple and low-skill-required mouse model that can simulate acute retinal ischemia and reperfusion injury in RAO patients.

To better simulate the retinal ischemic process and possible IRI following RAO, we developed a novel vascular-associated mouse model called the unilateral pterygopalatine ophthalmic artery occlusion (UPOAO) model. In this model, we employed silicone wire embolization and carotid artery ligation to completely block the blood supply to the retina. We characterized the major classes of retinal neural cells and evaluated visual function following different durations of ischemia (30 minutes and 60 minutes) and reperfusion (3-days and 7-days) after UPOAO. Additionally, we utilized transcriptomics to investigate the transcriptional changes and elucidate the pathophysiological process of the UPOAO model after ischemia and reperfusion. Finally, we highlighted the transcriptomic differences between this model and other retinal ischemia-reperfusion models, including HIOP and UCCAO, revealing the unique pathological processes that closely resemble retinal IRI in RAO. The UPOAO model offers new insights into the mechanisms and pathways involved in ischemia and reperfusion studies of RAO, providing a foundation for studying protective strategies for ocular ischemic diseases.

2 Methods and materials

2.1 Animals

Eight-week-old male C57BL/6 mice were used in the experiments. Only male mice were used to exclude the potential influence of oestrogenic hormones. The mice were provided sufficient food and water and were maintained on a 12-hour dark-light cycle in a room with regulated temperature conditions. All the experimental procedures were designed and conducted according to the ethical guidelines outlined in the Association for Research in Vision and Ophthalmology (ARVO) Statement for the Use of Animals in Ophthalmic and Vision Research. The study protocol and methods received approval from the Experimental Animal Ethics Committee of Renmin Hospital of Wuhan University (approval number: WDRM-20220305A).

2.2 Preparation and surgical procedure of the UPOAO model

Eight-week-old male C57BL/6 mice weighing 20-25 grams were used in this study. An isoflurane-based anaesthesia system was used to induce and maintain general anaesthesia in the mice during surgical procedures. The mice were anaesthetized with a 1.5-2% concentration of isoflurane delivered in a mixture of nitrous oxide and oxygen via rubber tubing. Body temperature was maintained at 37 ± 0.5 °C throughout the surgery. Surgical instruments were sterilized with 75% ethanol before use to ensure sterility.

We drew inspiration from the widely employed middle cerebral artery occlusion (MCAO) model [18, 19], commonly used in cerebral ischemic injury research, which guided the development of the UPOAO model. The mouse was gently positioned in a supine posture on a heating blanket, and its neck was exposed. The preparation involved depilation of the neck area, followed by skin disinfection, and finally, a midline incision along the neck was made. After separating the neck gland using two tweezers, careful blunt dissection was employed to separate the left CCA, internal carotid artery (ICA) and external carotid artery (ECA), avoiding the compression of nearby nerves and veins (**Fig. 1A**) (**Video 1**). Then, the CCA and ICA were ligated with a 6-0 suture. A knot was secured using a 6-0 suture at the distal end of the ECA, and a slipknot was created at the proximal end (**Fig. 1B**). To preserve the reperfusion process and avoid bleeding and mortality during surgery, a silicone wire embolus was inserted through the ECA instead of through the CCA. Ophthalmic scissors were used to make a small inverted "V"-shaped incision between the two suture knots on the ECA. A specialized silicone wire embolus, measuring 7 ± 0.1 mm in length and 0.21 ± 0.1 mm in diameter, was inserted through the incision, directing it into the ECA and further

into the CCA (**Fig. 1C**). Next, the ligation on the ICA was removed, and the ECA was cut at the point where the artery had been previously incised with scissors. Subsequently, the silicone wire embolus was retracted to the bifurcation of the CCA, rotated counterclockwise, and inserted into the ICA and further into the pterygopalatine artery (PPA) (**Fig. 1D1**). The silicone tail of the silicone wire embolus was positioned near the bifurcation of the CCA (**Fig. 1D2**), effectively obstructing the ophthalmic artery (OA) (**Video 2**). Then, the slipknot was fastened, and the skin was sutured. The mouse was free to move around during arterial embolization.

Following a predetermined ischemia embolization time, the silicone wire embolus was carefully removed from the PPA without massive haemorrhage (**Fig. 1E**). The CCA suture was removed, restoring arterial reperfusion (**Fig. 1F**) (**Video 3**). Then, the skin wound was sutured, and the mice were routinely fed during reperfusion (**Fig. 1G**).

2.3 Staining of retinal vessels

To clearly visualize the retinal vessels, rhodamine-labelled canavalin A was used for mouse heart perfusion. Each mouse was anaesthetized with 1% sodium pentobarbital and placed in the supine position on a foam board resting on a tray. The sternum was opened to expose the heart. A needle attached to the perfusion tube was inserted into the left ventricle via the apex of the mouse heart and secured. The right atrial appendage was dissected with scissors to allow blood outflow. Saline solution was perfused through the left ventricle using a pump system to drain blood from the arteries. Subsequently, the prepared solution of rhodamine-labelled canavalin A was injected into the mice and circulated throughout the mouse vessels for 30 minutes. The mice were then sacrificed, and their eyes were removed and immersed in paraformaldehyde (PFA) for one hour in the dark. A flat-mounted retina was obtained, and retinal images were captured using a Leica SP8 confocal microscope equipped with a 10× objective.

2.4 Quantification of RGCs and microglia

The mice were euthanized by cervical dislocation, and their eyes were fixed in 4% PFA for 60 minutes at room temperature (RT). Subsequently, the cornea and lens were excised, and the intact retina was isolated for retinal flat mounting. The retina was immersed in PBS supplemented with 5% bovine serum albumin (BSA) and 0.5% Triton X-100 for overnight blocking and then incubated with Brn3a antibody (Santa Cruz Biotechnology, Dallas, Texas, USA) or Iba1 antibody (Wako, Japan). Following a two-day incubation with the primary antibody, the retina was gently washed three times with PBS. Subsequently, the retina was incubated with Alexa Fluor 594 (AntGene, Wuhan, China) in a light-protected cassette for two days. After three rinses, the retina was uniformly sectioned into a four-leaf clover morphology and flattened using a coverslip. Finally, retinal flat mounts labelled with Brn3a were imaged using a Leica SP8 confocal microscope (Leica TCS SP8, Germany) equipped with a 20 × objective, while those labelled with Iba1 were imaged utilizing a fluorescence microscope (BX63; Olympus, Tokyo, Japan). Each quadrant was consistently subdivided into central, middle and peripheral fields at regular intervals. The surviving RGCs and activated microglia were quantified and averaged using ImageJ software (National Institutes of Health, USA).

2.5 Electroretinogram (ERG)

Mice were dark-adapted overnight before ERG, and all subsequent procedures were carried out in the dark. Before ERG, the mice were anaesthetized with 1% sodium pentobarbital via intraperitoneal injection, and their pupils were dilated. A subcutaneous electrode was inserted into the posterior cervical skin, a tail electrode was affixed to the posterior end of the mouse tail, and the corneal electrode was gently placed on the central corneal surface. A RetiMINER-C visual electrophysiological system (3VMED Co., Ltd., Shanghai, China) was used for recording electrical

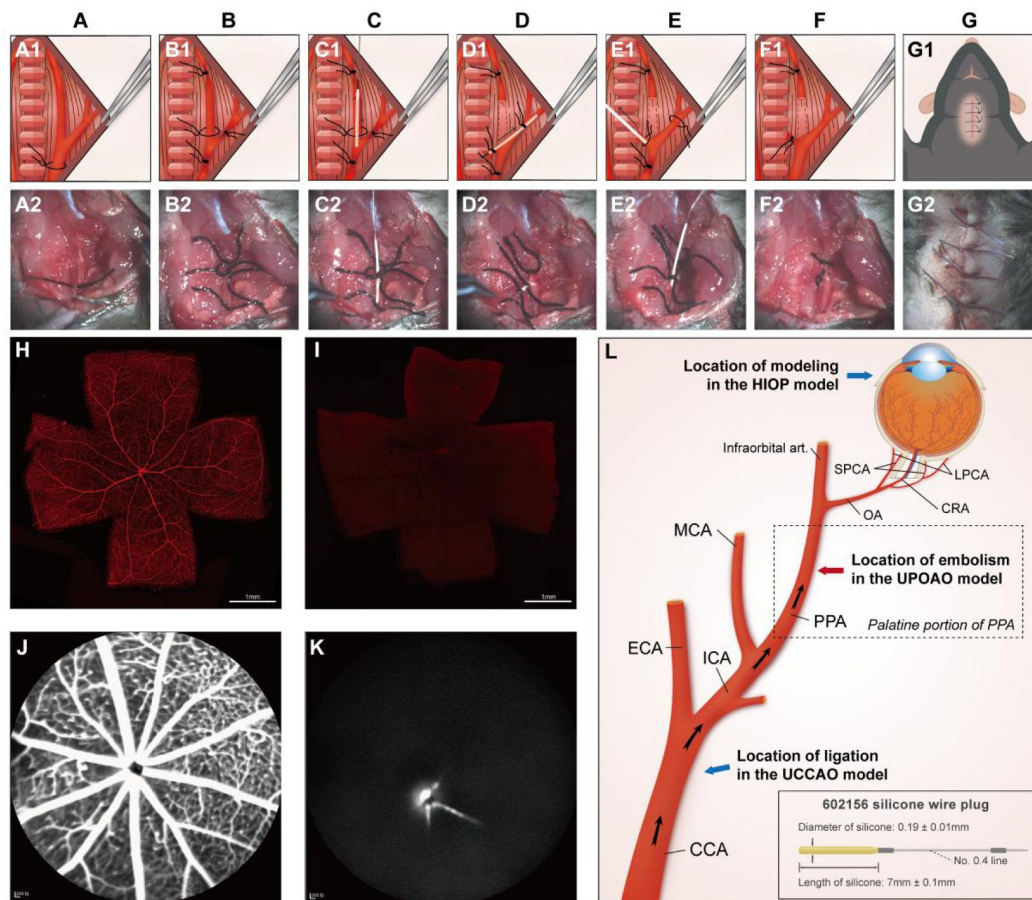


Fig. 1

Modeling Procedure, Validation, and Cervical Artery Anatomy

(A1-G1) Schematic illustration of unilateral pterygopalatine ophthalmic artery occlusion (UPOAO). (A2-G2) Practical operation of UPOAO. (A1, A2) Blunt separation and exposure of the left cervical arteries. (B1, B2) Arterial suture ligation. (C1, C2) Insertion of the silicone wire embolus. The artery was incised to create a hole, and the silicone wire embolus was inserted. (D1, D2) Artery disconnection and movement of the silicone wire embolus. The artery was cut along the incision, and the silicone wire embolus was retracted and reinserted. (E1, E2) Removal of the silicone wire embolus and reperfusion. (F1, F2) Suture removal. The sutures at both ends of the disconnected vessel were knotted, and the other two sutures were removed. (G1, G2) Anatomic reduction and suturing of the skin. (H, I) Canavalin A label vasculature of the UPOAO mouse retina. The silicone wire embolus was inserted into artery before perfusing rhodamine-labeled canavalin A into the heart of UPOAO mouse. The sham eye served as an unpracticed control eye, while the UPOAO lateral eye represented the experimental eye. Retinal vessels in the sham eye (H) exhibited fluorescence filling, while retinal vessels in the UPOAO lateral eye (I) remained unfilled. Scale bar = 1 mm. (J and K) Fluorescein Fundus Angiography (FFA) performed before removing the silicone wire embolus from the UPOAO mouse. The vessels in the sham lateral retina (J) were perfused, while the lateral retinal perfusion in UPOAO (K) was delayed. (L) Schematic illustration of cervical artery anatomy and ocular blood supply. Embolization of the PPA resulted in ocular ischemia. The red arrow indicates the site of the silicone wire embolus occlusion. The silicone wire embolus used a type 602156 wire, extended to 7mm with a diameter of 0.21mm. The blue arrows indicate the modelling locations of the HIOP model and the UCCAO model, respectively. CCA: common carotid artery; ICA: internal carotid artery; ECA: external carotid artery; PPA: pterygopalatine artery; MCA: middle cerebral artery; Infraorbital art.: infraorbital artery; OA: ophthalmic artery; SPCA: short posterior ciliary artery; LPCA: long posterior ciliary artery; CRA: central retinal artery.

responses. The a-waves, b-waves, and oscillatory potentials (OPs) after flash stimuli of 0.01, 0.03, 0.1, 0.3, 1.0, 3.0, and 10.0 cd.s/m² in scotopic adaptation were recorded. The amplitudes and implicit times of a-waves, b-waves, and OPs in response to various flash stimuli were analysed.

2.6 Optical Coherence Tomography Imaging (OCT) and Fluorescein Fundus Angiography (FFA)

The mice were placed on the platform of a Spectralis HRA + OCT device (Heidelberg Engineering, Heidelberg, Germany) for OCT imaging following anaesthesia. Pupil dilation was performed, and normal saline was applied regularly to maintain corneal moisture. The focal length of the device was adjusted until the mouse retina was clearly visible. The head of the mouse was gently repositioned to capture images of the peripheral fundus. The multiline mode was used to scan each layer of the retina, and four quadrants of view centred on the nipple in the upper left, lower left, upper right, and lower right were recorded. In this study, the total retinal thickness was manually segmented into three parts and encompassed the entire thickness from the nerve fibre layer to the photoreceptor layer. The ganglion cell complex (GCC) was defined as the combined thickness of the retinal nerve fibre layer (RNFL), ganglion cell layer (GCL), and inner plexiform layer (IPL). The inner nuclear layer (INL) and the outer plexiform layer (OPL) were combined for thickness analysis due to the difficulty in distinguishing between these layers. The remaining retinal layers, including the outer nuclear layer (ONL), inner segment/outer segment (IS/OS), and retinal pigment epithelium (RPE) layers, were measured together. The thickness of the retina at 1.5 papillary diameters (PD), 3.0 PD, and 4.5 PD from the centre of the optic disc was measured using the Heidelberg measuring tool.

For FFA, the mice were anaesthetized, and their pupils were dilated. Subsequently, fundus angiography was performed immediately following an intraperitoneal injection of sodium fluorescein. Images were acquired alternately for both eyes within 5 seconds of the injection.

2.7 Hematoxylin and eosin (HE)

Mouse eyes were fixed in FAS eyeball fixation solution (Service-bio, Wuhan, China) for 48 hours, followed by dehydration and subsequent embedding in paraffin. The paraffin blocks were trimmed parallel to the optic nerve to obtain 3–4 µm thick sections where the optic nerve was located. Six paraffin sections from each eyeball were stained with HE. Images were captured using an Olympus fluorescence microscope. The retinal thickness within 500 µm of the optic disc was measured using Image-Pro Plus 6.0 software.

2.8 Staining of Retinal Sections

For immunofluorescence, mouse eyes were fixed in 4% PFA solution and subjected to gradient dehydration using 10%, 20%, and 30% sucrose solutions. The following day, the eyes were embedded in optimum cutting temperature (OCT) compound (SAKURA, USA) and frozen at -80 °C. Several 14-µm-thick frozen sections were obtained from each eyeball through the use of a freezing microtome (Leica, Wetzlar, Germany). The section surface was parallel to the optic nerve, and 3–6 frozen sections were mounted on a single slide. The slides were stored at -20°C until use.

All sections were blocked with 5% BSA and 0.5% Triton X-100 in PBS for 2 hours. Primary antibodies, as listed in **Table 1**, were diluted in PBS containing 5% BSA and 0.5% Triton X-100 and incubated at 4 °C overnight. The following day, after three rinses, the sections were incubated with secondary antibodies at RT for 2 hours in a cassette, followed by three washes. The sections were then stained with DAPI (Service-bio, Wuhan, China) for 15 minutes. After the final three rinses, the frozen sections were sealed. A Leica SP8 instrument equipped with a 40× objective was used to photograph frozen sections near the optic nerve head. The fluorescence intensity of all slices of each eye was determined using ImageJ software.

Antibodies	Species	Dilution	Company
PKC α	rabbit	1:1000	Sigma
Recoverin	rabbit	1:1000	Millipore
Calbindin	rabbit	1:5000	SWANT
ChAT	goat	1:200	Millipore
Iba1	rabbit	1:500	Wako
CD45	rabbit	1:100	CST
Rhodamine-labeled canavalin A	/	1:125	MKBio
Goat anti-mouse IgG	goat	1:500	Jackson
Alexa Fluor 488			
Donkey anti-rabbit	donkey	1:500	Jackson
IgG Alexa Fluor 594			
Donkey anti-goat IgG	donkey	1:500	Jackson
Alexa Fluor 488			

Table 1

Antibodies Used in Staining of Flat-Mounted Retina and Sections

2.9 Surgical procedure for the UCCAO model and the HIOP model

Animals for the UCCAO model were prepared and managed according to the procedure described for the UPOAO model. After blunt dissection, the left CCA was ligated with a 6-0 suture. After 60 minutes of ligation, the suture was removed, and the skin wound was subsequently closed with sutures. Mice were routinely fed during the reperfusion period, and the retinas of UCCAO mice were harvested 7-days post-surgery.

The HIOP model was generated based on previously established methods [20]. Briefly, mice were anaesthetized by intraperitoneal injection of 1% sodium pentobarbital. Pupils were dilated using tropicamide phenylephrine eye drops (0.5% tropicamide and 0.5% deoxyadrenaline hydrochloride, Santen Pharmaceutical) applied 5 minutes in advance. An insulin hypodermic needle attached to a silicone elastomer tube was inserted into the anterior chamber of the unilateral experimental eye. The IOP was elevated for 60 minutes by raising the height of the saline solution storage bag to 162 cm, after which the needle was removed to immediately restore the IOP to baseline levels. Iris whitening and loss of retinal red-light reflexes indicated retinal ischemia. After surgery, levofloxacin hydrochloride ophthalmic gel was applied to the eyes of the mice. Retinas from HIOP mice were collected 7-days post-surgery.

2.10 Transcriptome sequence and analysis

Retinal samples from UPOAO, HIOP, and UCCAO mice were extracted and promptly frozen in liquid nitrogen within 10 minutes of cervical dislocation. Total RNA extraction was performed according to the protocol outlined in the TRIzol Reagent manual (Life Technologies, CA, USA). RNA integrity and concentration were assessed using an Agilent 2,100 Bioanalyzer (Agilent Technologies, Inc., Santa Clara, CA, USA). The resulting RNA samples were then pooled and subjected to sequencing on the Illumina HiSeq3000 platform in a 150-bp paired-end read format. The raw RNA-sequencing (RNA-seq) reads were preprocessed and quantified using the featureCounts function in the SubReads package version 1.5.3, with default parameters.

RNA-seq analysis was conducted on samples from 5 mice in the non-perfusion group, 4 mice in the 3-days perfusion group, and 5 mice in the 7-days perfusion group of the UPOAO model. Additionally, retinas from both the 7-days HIOP experimental eyes and bilateral UCCAO eyes were collected (n=5) for analysis.

Data normalization and subsequent processing utilized the 'limma' package in R software (version 4.2.3) [21]. Volcano plots and Venn diagrams were generated for visualization to identify significantly differentially expressed genes (DEGs) and perform Gene Ontology (GO) annotation analysis. DEGs with a $|\log_2\text{-fold change (FC)}| > 1$ and adjusted p value < 0.05 were significantly differentially expressed. Enrichment analysis focused on DEGs with an adjusted p < 0.05 and an enriched gene count > 5 . To visualize the expression patterns of significant DEGs, a heatmap was generated using the 'pheatmap' package. Gene Set Enrichment Analysis (GSEA) was conducted using the GSEA program (version 4.3.2) [22], employing the default gene set m-subset (mh.all.v2023.1.mm.symbols.gmt) to explore significant functional and pathway differences. Enriched pathways were classified based on criteria such as adjusted p value (< 0.05), false discovery rate (FDR) q value (< 0.25), and normalized enrichment score ($|\text{NES}| > 1$). Furthermore, protein – protein interaction (PPI) analysis was performed using the Search Tool for the Retrieval of Interacting Genes (STRING) database (<https://string-db.org/>) [23]. Genes with a greater degree of protein-level interactions with others were further analysed using Cytoscape software (version 3.8.2) to generate a downstream PPI map. PPI pairings with an interaction score > 0.7 were extracted and visualized using Cytoscape 3.9.0 [24].

2.11 Real-time Quantitative polymerase chain reaction (RT-qPCR)

RNA was extracted from fresh mouse retinal samples and subjected to reverse transcription using HiScript III RT SuperMix for qPCR (Vazyme, China) following the manufacturer's instructions. RT-qPCR analysis was conducted on a CFX instrument (Bio-Rad) using AceQ qPCR SYBR Green Master Mix (Vazyme, China). The PCR program comprised 40 cycles of 10 seconds at 95°C and 30 seconds at 60°C. Assays were performed in triplicate, and Ct values were normalized to beta-actin levels. Relative quantification of target gene expression was calculated using the $2^{-\Delta\Delta C_t}$ method. The primer sequences are provided in [Table S1](#).

2.12 Statistical Analysis

All the data were analysed using GraphPad Prism version 9.0 (GraphPad Software, San Diego, CA, USA). Each experimental group included a minimum of three biological replicates. T tests, including RGCs and microglia counting, OPs analysis, retinal thickness measurements via HE staining, and fluorescence intensity quantification, were used to compare the two groups. Two-way ANOVA (two-way analysis of variance) test was used for more than two different experimental groups, including ERG waves and retinal thickness measurements in OCT. All data are presented as the mean $\pm$ standard error of the mean (SEM), and the statistical graphs are shown as scatter bar graphs and line charts. $P < 0.05$ was considered to indicate statistical significance.

3 Results

3.1 Silicone wire embolus insertion interrupts retinal blood flow

To reproduce the retinal ischemic process and potential IRI in RAO, we established a mouse model of UPOAO by combining silicone wire embolization with carotid artery ligation. Model validation was conducted through a series of experiments, with detailed descriptions provided in the Materials and Methods section ([Fig. 1A-G](#)). To confirm blood flow disruption, we performed cardiac perfusion with fluorescently labelled lectin and FFA. Rhodamine-labelled canavalin A was used during in vivo perfusion with silicone wire embolus, and both eyes were evaluated. The sham eye exhibited normal blood perfusion in the retina, while the experimental eye showed no perfusion ([Fig. 1H, I](#)). FFA revealed delayed and limited perfusion in the experimental lateral retina, primarily near the optic disc ([Fig. 1J, K](#)). These findings indicate that the insertion of the silicone wire embolus effectively impaired blood flow to the retina.

3.2 60-minute ischemia in UPOAO damage retinal structure and function

To investigate the optimal ischemic duration, the mice were subjected to ischemic periods of 30 and 60 minutes, followed by reperfusion periods of 3-days and 7-days. Retinal structure and visual function were assessed through flat-mounted retina analysis and flash ERG.

In the 30-minute ischemia group, no significant RGCs death was observed after 3-days ([Fig. 2A, B](#)) or 7-days of reperfusion ([Fig. 2C, D](#)). However, after 60 minutes of ischemia followed by either 3-days or 7-days of reperfusion, reductions in RGCs density were evident ([Fig. 2E-H](#)). ERG results showed no statistical difference in amplitudes between bilateral eyes after 3- and 7-days of 30-minutes reperfusion ([Fig. 3A-C](#)). However, the b-wave amplitude notably decreased after 60 minutes of ischemia and 3-days of reperfusion ([Fig. 3D](#)). By 7-days of reperfusion, the b-wave amplitude had halved compared to that of the sham eyes ([Fig. 3E](#)). The appearance times of the a-waves and b-waves in the experimental and control eyes remained consistent across all four groups ([Fig. S1](#)). Additionally, OPs have also been used to sensitively monitor retinal ischemic

effects by detecting changes before any alterations in b-waves occur [25–27]. The amplitudes of OPs in the experimental eyes, particularly in the 60-minute ischemia and 7-days reperfusion group, significantly decreased to less than 50% of those in the sham eyes (Fig. 3F, G). Based on RGCs survival and changes in ERG waves, we determined that a 60-minute ischemic duration is optimal for the UPOAO model.

3.3 Evaluation of retinal thickness in UPOAO through OCT and HE

To assess retinal thickness non-invasively in the UPOAO model, we conducted OCT imaging during 3-days and 7-days reperfusion periods. Following a 3-days reperfusion period, no significant changes were observed in the GCC, INL + OPL, and ONL + IS/OS + RPE layers at 1.5, 3, and 4.5 PD from the optic disc (Fig. 4B, Fig. S2A, B). However, the total retinal thickness decreased at 4.5 PD (Fig. 4C). After 7-days of reperfusion, the total retinal thickness decreased at 1.5 PD, 3 PD, and 4.5 PD, primarily due to inner retinal thinning (Fig. 4D–F, Fig. S2C, D).

For a detailed analysis of inner retinal thickness, we extracted UPOAO mouse eyes and stained them with HE. The thickness of the RNFL + GCL increased at 3-days of reperfusion (Fig. 4H), followed by a decrease at 7-days of reperfusion (Fig. 4L). IPL thickness decreased at both 3-days and 7-days of reperfusion (Fig. 4I, M), while INL thickness decreased only at 7-days of reperfusion (Fig. 4N). We hypothesized that the initial swelling of the GCL in response to acute ischemia and hypoxia led to early thickening, followed by gradual thinning due to tissue dysfunction. This hypothesis was supported by the observation of RGCs loss and reduced GCC thickness evaluated by OCT at 7-days (Fig. 4E).

We observed by OCT that changes in retinal thickness became noticeable at 3-days post-UPOAO and exhibited a significant decrease at 7-days. HE results further revealed that alterations in inner retinal thickness constituted a substantial portion of the total retinal thickness during the early reperfusion stage after UPOAO.

3.4 Survival of retinal neural cells in UPOAO

Given that mouse retinal function heavily relies on rod cells and that scotopic (low light) vision is primarily governed by rod photoreceptors, the observed reductions in b-wave and OPs-wave amplitudes indicate impaired synaptic transmission within the inner retina [28]. Our OCT and HE findings suggested structural injuries within the inner retina. Notably, the thickness of the outer retinal layers in OCT remained unchanged. To further explore these observations, we conducted immunofluorescence staining to investigate alterations in major retinal cell types, primarily focusing on bipolar cells (BCs), photoreceptor cells, horizontal cells (HCs), and cholinergic amacrine cells.

3.4.1 BCs loss and photoreceptor cells survival

PKC α serves as a marker delineating BCs, with their cell bodies primarily located in the outermost part of the INL, axonal terminals extending into the innermost part of the IPL, and dendrites confined to the OPL (Fig. 5A, B). In the experimental UPOAO eyes, BCs did not show significant changes at the 3-days reperfusion mark but exhibited dramatic alterations by the 7-days reperfusion period (Fig. 5A2, B2). In particular, the immunostaining density of BCs dendritic and axonal arbors notably decreased at the 7-days reperfusion mark. BCs account for approximately 40% of INL cells in mice, and their somata and axonal processes form a substantial part of the INL and IPL, consistent with the thinning of the INL observed in both OCT and HE at 7-days (Fig. 4E, N). Notably, the decrease in the number of PKC α^+ BCs at the 7-days reperfusion point coincided with the decrease in the b-waves amplitude in ERG (Fig. 3E), indicating functional interplay between BCs and other cells.

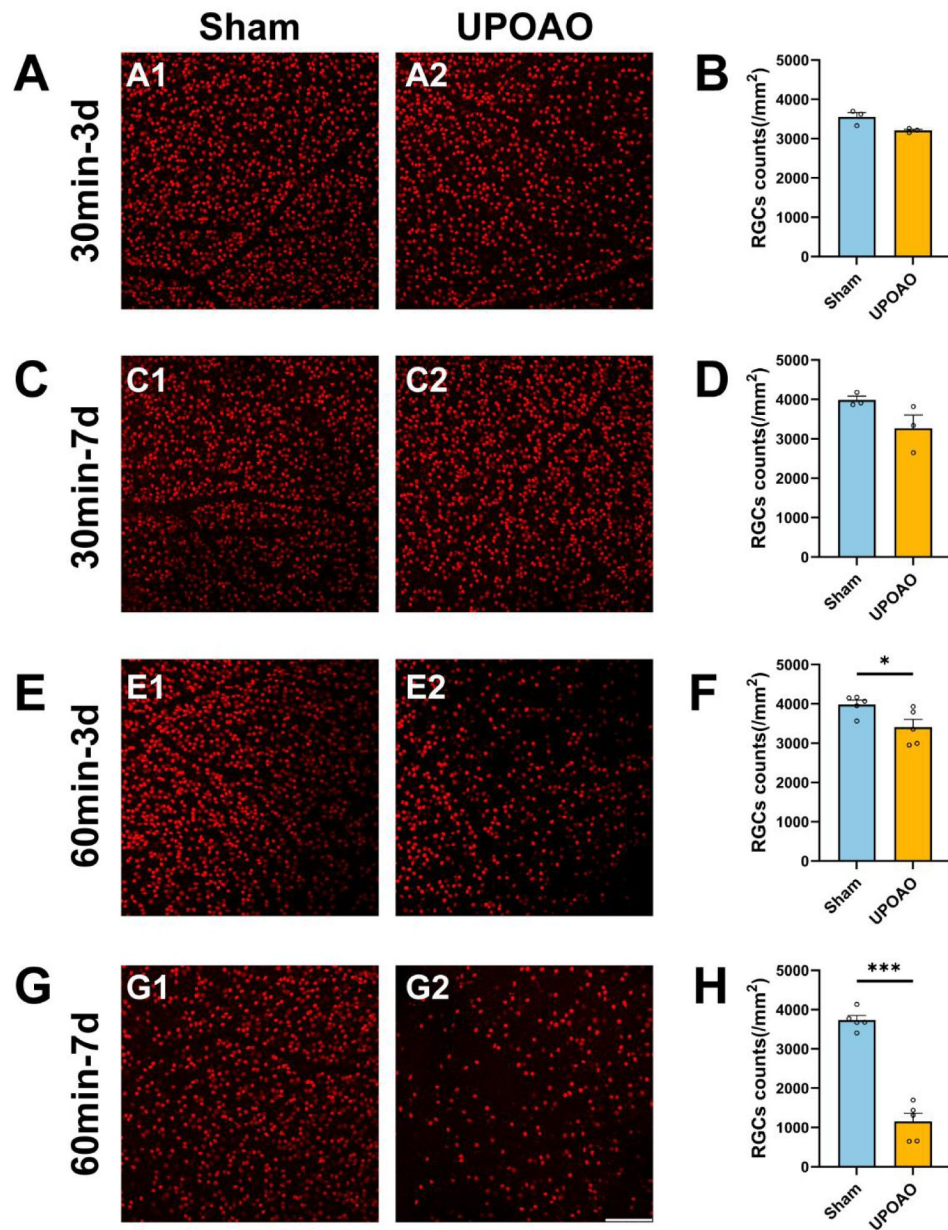


Fig. 2

Staining and Quantification of Retinal Ganglion Cells (RGCs) at Different Ischemia and Reperfusion Times.

Flat-mounted retina RGCs were labeled with Brn3a staining. (A, B) Representative pictures of the peripheral retinal visual field (A), and quantification of surviving RGCs (B) in the 30-minute ischemia and 3-days reperfusion group. $n = 3$. (C, D) Representative pictures of the peripheral retinal visual field (C), and quantification of surviving RGCs (D) in the 30-minute ischemia and 7-days reperfusion group. $n = 3$. (E, F) Representative pictures of the peripheral retinal visual field (E), and quantification of surviving RGCs (F) in the 60-minute ischemia and 3-days reperfusion group. $n = 5$. (G, H) Representative pictures of the peripheral retinal visual field (G), and quantification of surviving RGCs (H) in the 60-minute ischemia and 7-days reperfusion group. $n = 5$. The results showed a significant loss of RGCs after 60 minutes of ischemia. Data were presented as means $\pm$ s.e.m, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$, t-test. Scale bar = 100 μ m.

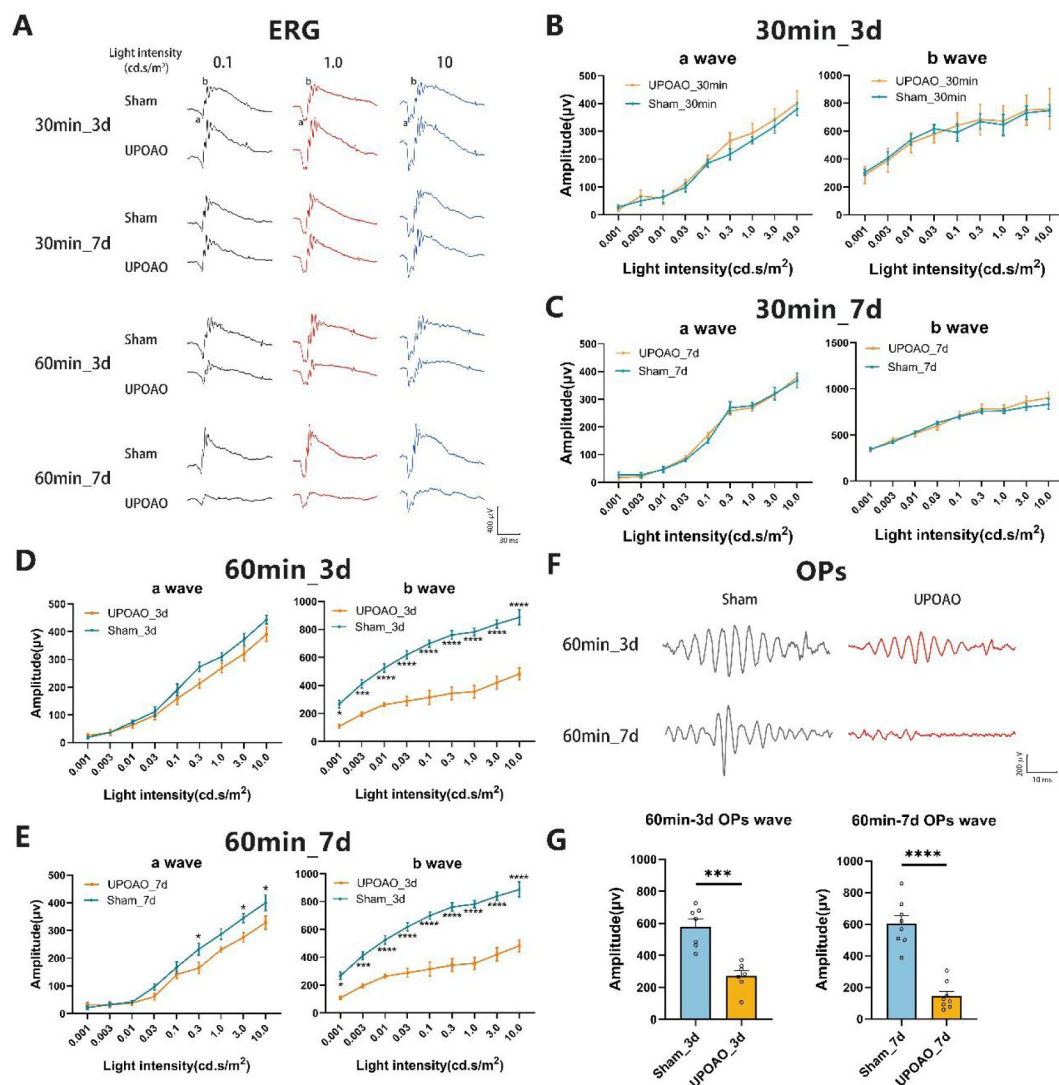


Fig. 3

Comparison of Electretinographic (ERG) Dark-Adapted Responses at Different Ischemia and Reperfusion Times.

Following the evaluation of surviving RGCs, visual function in sham and UPOAO experimental eyes at various ischemia and reperfusion times was assessed using ERG. (A) Representative waveforms in the four groups at the stimulus light intensities of 0.1, 1.0, and 10.0 cd.s/m², respectively. (B) Quantification of a-wave and b-wave amplitudes in the 30-minute ischemia and 3-days reperfusion group. $n = 5$. (C) Quantification of a-wave and b-wave amplitudes in the 30-minute ischemia and 7-days reperfusion group. $n = 5$. (D) Quantification of a-wave and b-wave amplitudes in the 60-minute ischemia and 3-days reperfusion group. $n = 7$. (E) Quantification of a-wave and b-wave amplitudes in the 60-minute ischemia and 7-days reperfusion group. $n = 8$. Dark-adapted responses showed almost similar a-wave amplitudes but significantly decreased b-wave amplitudes in the 60-minute ischemia groups. The amplitudes of b-waves declined at 3-days and even more prominently at 7-days. (F, G) Representative OPs and quantification of amplitudes in the 60-minute ischemic groups. $n = 7$ in the 3-days reperfusion group; $n = 8$ in the 7-days reperfusion group. The amplitudes of OPs decreased significantly at 7-days reperfusion. The decline in b-waves and OPs along with the loss of RGCs, supports the selection of a 60-minute ischemic duration as an appropriate choice. Data were presented as means $\pm$ s.e.m, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$, two-way ANOVA test for a-waves and b-waves; paired t-test for OPs.

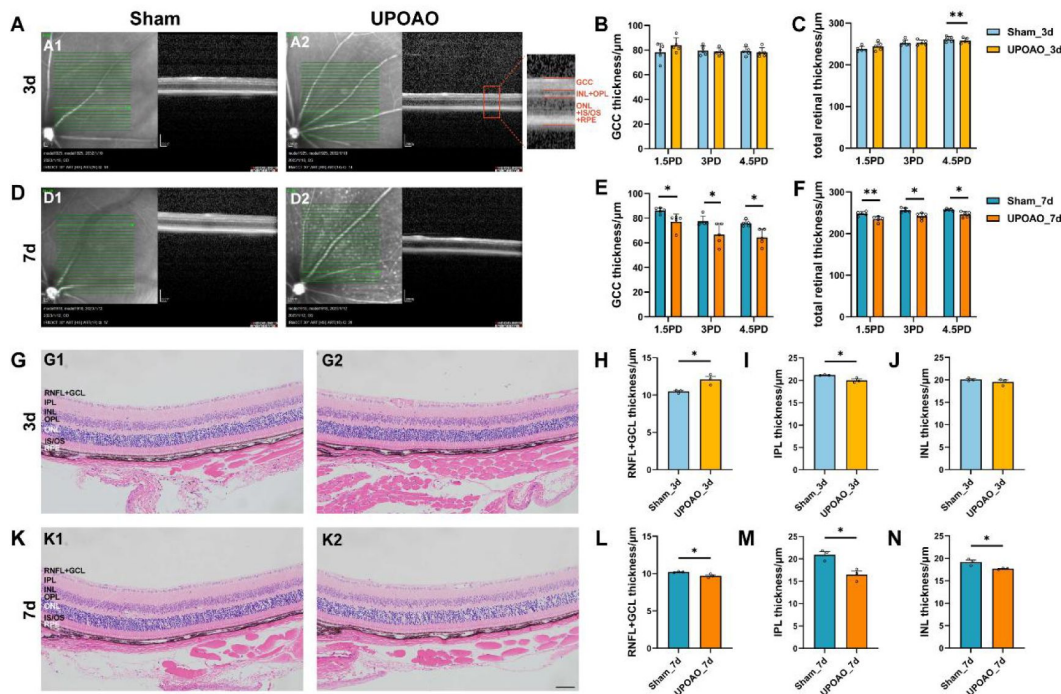


Fig. 4

Changes in Retina Morphology in the UPOAO Model.

(A) Representative OCT images of the mouse retina at 3-days. The Green lines indicate the OCT scan area starting from the optic disc. Local magnification and layering are annotated. GCC: ganglion cell complex, including RNFL, GCL and IPL layers. (B and C) Quantification of GCC and total retinal thickness at 3-days. The thickness of the GCC and the entire retina in OCT was measured and compared at distances of 1.5 PD, 3.0 PD and 4.5 PD from the optic disc, respectively. $n = 5$. (D) Representative OCT images of the mouse retina at 7-days. (E and F) Quantification of GCC and total retinal thickness at 7-days. $n = 5$. (G) Representative HE images of the mouse retina at 3-days. (H, I and J) Quantification of NFL + GCL, IPL, and INL thickness at 3-days. Retinal thickness in HE was measured near the optic nerve head and compared. $n = 3$. (K) Representative HE images of the mouse retina at 7-days. (L, M and N) Quantification of NFL + GCL, IPL, and INL thickness at 7-days. $n = 3$. Data were presented as means $\pm$ s.e.m, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$, two-way ANOVA test in OCT and paired t-test in HE. PD: papillary diameters; RNFL: retinal nerve fiber layer; GCL: ganglion cell layer; IPL: inner plexiform layer; INL: inner nuclear layer; OPL: outer plexiform layer; ONL: outer nuclear layer; IS: inner segment; OS: outer segment; RPE: retinal pigment epithelium. Scale bar = 50 μ m.

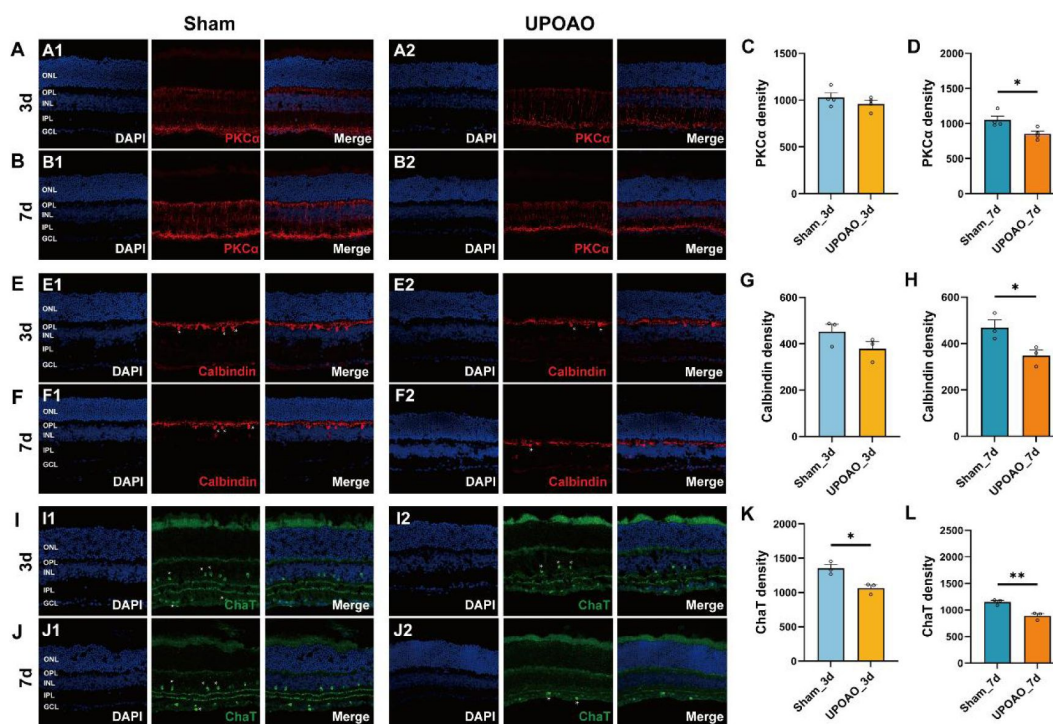


Fig. 5

Changes in Bipolar Cells, Horizontal Cells, and Cholinergic Amacrine Cells in UPOAO Mice.

(A) Representative images of mouse retina co-stained with DAPI and PKC α at 3-days (B) Representative images of mouse retina co-stained with DAPI and PKC α at 7-days. (C) Quantification of PKC α fluorescence density at 3-days. (D) Quantification of PKC α fluorescence density at 7-days. $n = 4$. (E) Representative images of mouse retina co-stained with DAPI and Calbindin at 3-days. (F) Representative images of mouse retina co-stained with DAPI and Calbindin at 7-days. Horizontal cell somata are indicated by white asterisks. (G) Quantification of Calbindin fluorescence density at 3-days. (H) Quantification of Calbindin fluorescence density at 7-days. $n = 3$. (I) Representative images of mouse retina co-stained with DAPI and ChAT at 3-days. (J) Representative images of mouse retina co-stained with DAPI and ChAT at 7-days. Cholinergic amacrine cell somata are indicated by white asterisks. (K) Quantification of ChAT fluorescence density at 3-days. (L) Quantification of ChAT fluorescence density at 7-days. $n = 3$. Data were presented as means $\pm$ s.e.m, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, t-test. Scale bar = 50 μ m.

Recoverin marks the somata and outer segments of photoreceptor cells and is localized within the ONL (**Fig. S3A, B**). Interestingly, recoverin-positive photoreceptors remained relatively stable throughout the reperfusion periods, which aligns with the unchanged thickness of the outer retinal layers observed in OCT (**Fig. S2C, D**).

3.4.2 HCs and cholinergic amacrine cells loss

We evaluated HCs and cholinergic amacrine cells using calbindin and ChAT immunostaining, respectively. Calbindin immunostaining highlighted the somata of HCs within the INL, with terminal axon connections linearly positioned within the OPL (**Fig. 5E, F**). In the UPOAO experimental eyes, no significant change in the number of HCs was observed during the 3-days reperfusion period, while a notable reduction was observed after 7 days. By 7-days, a significant decrease in cell body numbers, a considerable reduction in axon density, and disrupted linear connections were observed (**Fig. 5F2**).

ChAT⁺ cell bodies were located in the GCL and INL, with dendrites forming two narrow stratified bands within the IPL. The immunofluorescence density of ChAT⁺ amacrine cells decreased notably after 3 days and even more prominently after 7 days (**Fig. 5I, J**). The fluorescence intensity of the two bands within the IPL markedly decreased and was nearly invisible after 7 days (**Fig. 5J2**).

3.5 Time course transcriptome analysis revealed features of different reperfusion periods in UPOAO

To explore the pathophysiological processes of UPOAO, we extracted retinas from both eyes for transcriptome sequencing. The samples included retinas from eyes subjected to 60 minutes of ischemia without reperfusion, 60 minutes of ischemia followed by 3-days of reperfusion, and 60 minutes of ischemia followed by 7-days of reperfusion, with the sham-treated eyes serving as controls.

In the non-perfusion group, 215 genes were upregulated and 204 genes were downregulated (**Fig. 6A**). GO enrichment analysis revealed that the DEGs were related to leukocyte migration, epidermis development, myeloid leukocyte migration and other cell migration pathways (**Fig. 6B**). The heatmap and box showed the upregulated and downregulated genes involved in immune cell migration-related pathways during the non-perfusion period in UPOAO (**Fig. 6C**). High-connectivity DEGs ('hub genes'), such as *Dusp1* and *Fos*, were also enriched in these pathways (**Fig. S4A**). GSEA also showed similar results (**Fig. S4B-D**). These results suggested that in the early stage of retinal ischemic injury, leukocytes from the microvasculature may infiltrate retinal tissue. More experimental validation will be performed to confirm this hypothesis.

In the 3-days reperfusion group, 372 genes were upregulated, and 170 genes were downregulated (**Fig. 6D**). GO enrichment analysis revealed that the DEGs at 3-days after reperfusion were related to energy metabolism, mitochondrial regulation, and oxidative stress pathways (**Fig. 6E**). The heatmap and box displayed the upregulated and downregulated genes involved in oxidative stress-related pathways at the 3-days reperfusion stage in UPOAO, including hub genes such as *Mrpl18* and *Mrps23* (**Fig. 6F**, **Fig. S5A**). GSEA revealed changes in pathways related to the binding membrane of organelles, the cell surface and the plasma membrane protein complex (**Fig. S5B**). Furthermore, 44 overlapping genes between the mitochondrial genes and DEGs at 3-days post-UPOAO were mainly related to mitochondrial transport and the mitochondrial tricarboxylic acid cycle (**Fig. S5C, D**). These results suggested that mitochondria and metabolism play significant roles in IRI.

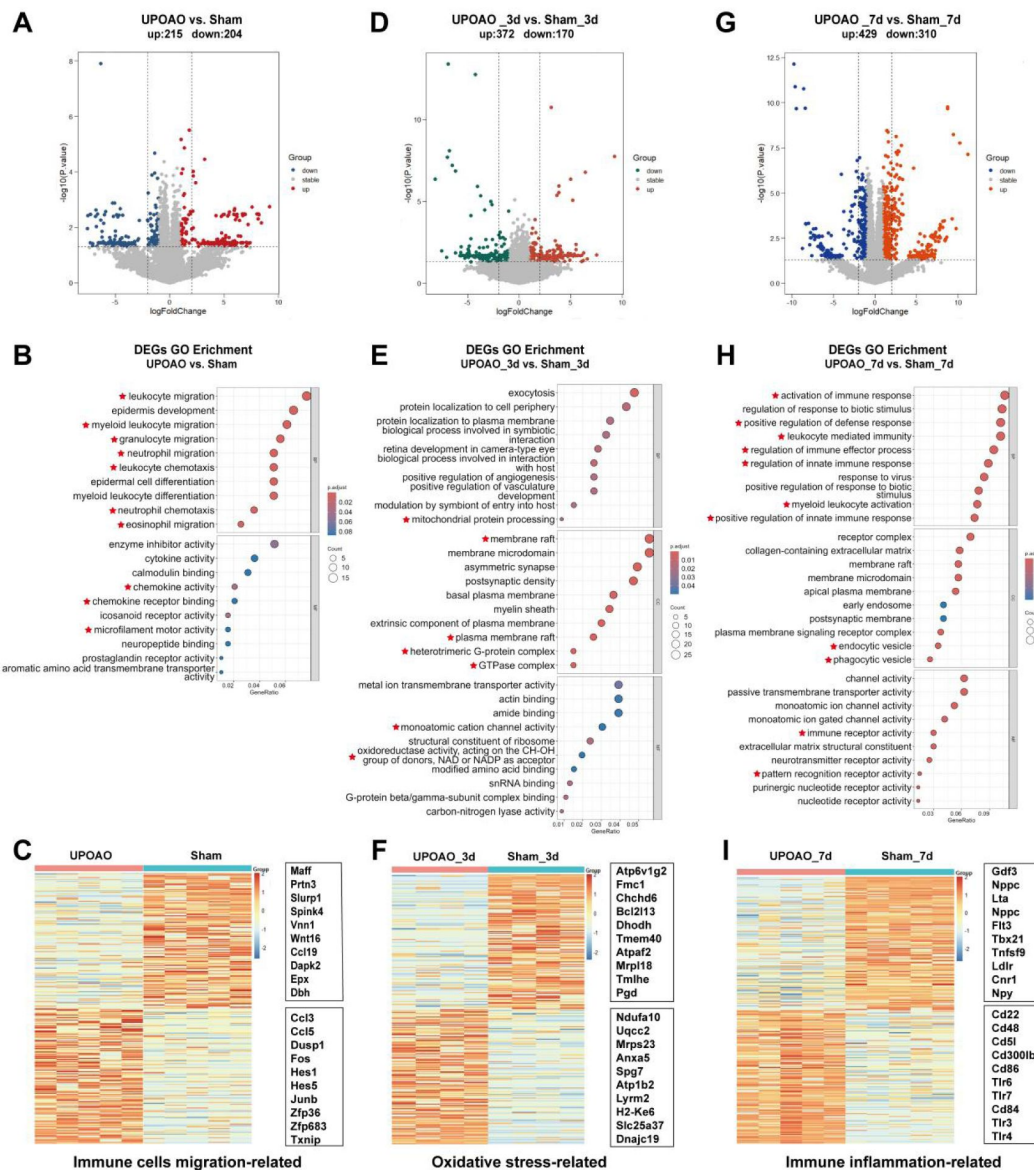


Fig. 6

Transcriptomic Features at Different Times of Reperfusion.

RNA-seq evaluation was performed at 0-day, 3-days, and 7-days reperfusion periods in UPOAO revealing enrichment in pathways related to immune cells migration, oxidative stress, and immune inflammation. (A, D, G) Volcano plots display differential expression genes (DEGs) between UPOAO and sham eyes in the no-perfusion group (A), 3-days perfusion group (D), and 7-days perfusion group (G), respectively. Red dots: significant upregulated genes, green and blue dots: significant downregulated genes, grey dots: stable expressed genes, adjusted $P < 0.05$. $\log_2FC = 1$. (B, E, H) Gene ontology (GO) analysis of differential genes in the non-perfusion group (B), 3-days group (E), and 7-days group (H). The DEGs between UPOAO and sham eyes were enriched in pathways associated with immune cells migration (0d), oxidative stress (3d), and immune inflammation (7d), respectively. (C, F, I) Heatmap displaying the top 100 downregulated and top 100 upregulated DEGs between UPOAO and sham at non-perfusion, 3-days, and 7-days reperfusion, respectively. The box represents the genes related to immune cells migration (C), oxidative stress (F), and immune inflammation (I). The ranking was determined by the magnitude of fold change. In each heatmap, the upper box represents the top 10 downregulated genes, while the lower box represents the top 10 upregulated genes.

In the 7-days reperfusion group, 429 genes were upregulated, and 310 genes were downregulated (Fig 6G). The DEGs at this stage were enriched in pathways related to immune effector processes, positive regulation of responses to external stimuli and inflammation (Fig. 6H). A heatmap was generated to show the upregulated and downregulated genes of immune inflammation-related pathways at 7-days after reperfusion in UPOAO, and most hub genes, such as Cd86, Cd48, Tlr4, and Tlr6 were also enriched in these pathways (Fig. 6I, Fig. S6A). GSEA also showed similar results (Fig S6B). Moreover, RT – qPCR confirmed the upregulation of immune inflammation-related gene expression (Fig. S7). Significantly, 112 genes that overlapped between immune-related genes and DEGs at 7-days post-UPOAO were primarily associated with immune receptor activity and phagocytic vesicles (Fig. S6C, D). These results underscore the predominant role of immune responses during this stage.

To explore the associations between changes observed at 3-days and 7-days after reperfusion, we analyzed the DEGs and identified 17 overlapping genes (Fig. S8A), which were mainly associated with the nitric oxide biosynthetic process (Fig. S8B). Our findings suggest that the immune inflammatory response observed after 7-days of reperfusion may represent the cumulative effect of the acute oxidative stress response during the initial 3-days reperfusion period.

3.6 Leukocyte Infiltration and Microglial Activation

Time course transcriptome analysis underscored the predominant role of immune inflammatory responses during the reperfusion stage. Previous reports have indicated that the inflammatory response in retinal ischemia-reperfusion injury (RIRI) is orchestrated by peripheral immune cells and resident immune cells within the retina [29, 30]. In our study, we demonstrated the presence of leukocyte infiltration and microglial activation in the UPOAO model. Immunofluorescence staining for CD45 was performed on retinas at different time points post modelling to visualize the distribution and quantity of white blood cells (Fig. 7A-D). Laser confocal Z-plane projections confirmed minimal leukocyte infiltration in the retinas of sham eyes, which showed that CD45⁺ cells in the vascular lumen likely represented patrolling cells (Fig. 7A). A significant increase in CD45⁺ cells was observed in UPOAO model retinas after 1 day (Fig. 7E), with a progressive increase in quantity over time (Fig. 7F, G). Notably, the majority of CD45⁺ cells in UPOAO model retinas at 1-day displayed a morphology similar to that of vascular leukocytes (Fig. 7B), suggesting a large influx of peripheral white blood cells into the retinal tissue. In contrast, CD45⁺ cells at 3-days and 7-days of reperfusion exhibited a combination of amoeboid and branched morphologies (Fig. 7C, D), indicative of activated states.

Microglia, which are known to undergo activation and morphological changes in response to various insults, were examined in the superficial retinas of sham and UPOAO eyes. Immunofluorescence staining for Iba1 was conducted on flat-mounted retinas at 3-days and 7-days post-reperfusion. In sham eyes, Iba1⁺ microglia displayed small somas and elongated dendrites that were evenly distributed within the retina (Fig. 7H1, I1). Conversely, UPOAO retinas exhibited numerous activated Iba1⁺ microglia with enlarged somas, shortened dendrites, and an amoeboid appearance (Fig. 7H2, I2). The number of Iba1⁺ microglia in UPOAO retinas was approximately five times greater than that in sham eyes at 3-days (Fig. 7J), with a further increase observed at 7-days (Fig. 7K), although this increase was less pronounced than that at the 3-days time point (Fig. 7J). These results indicated a potential reduction in immune-mediated inflammation over time.

3.7 RNA-seq comparison between UPOAO and extravascular occlusion models

To further study the characteristics of UPOAO, we analyzed the transcriptomes of two extravascular occlusion models: HIOP and UCCAO. The HIOP model induces ischemia through anterior chamber perfusion with normal saline (Fig. 8A), while the UCCAO model involves

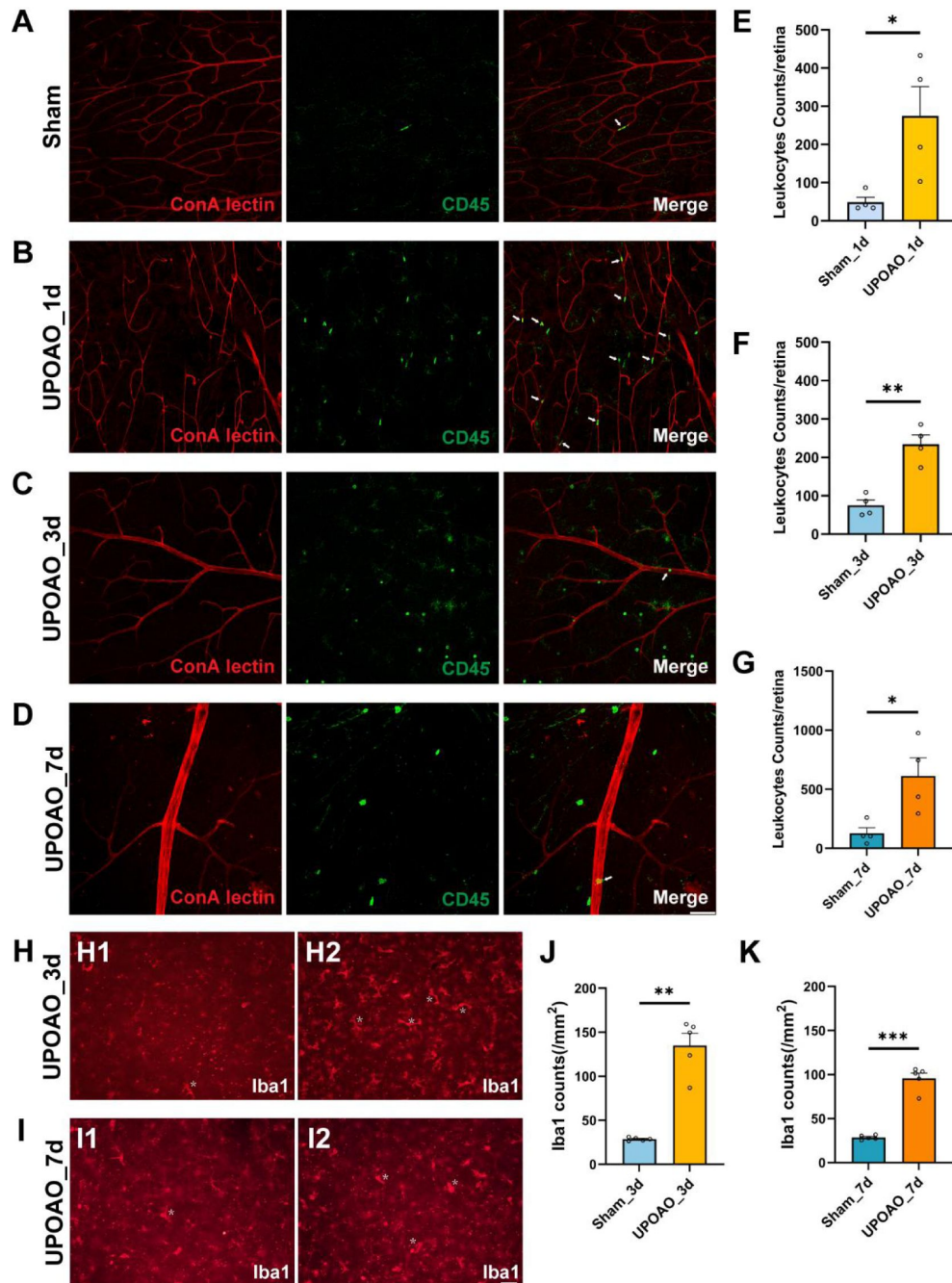


Fig. 7

Peripheral Leukocyte Infiltration and Retinal Resident Microglial Activation.

Rhodamine-labeled canavalin A was used for immediate cardiac perfusion to visualize blood vessels, followed by CD45 immunofluorescent staining to observe the relationship between blood vessels and CD45⁺ cells in sham (A), 1-day perfusion group (B), 3-days perfusion group (C), and 7-days perfusion group (D). The presence of CD45⁺ cells within blood vessels is indicated by white arrows. Scale bar = 50 μ m. CD45⁺ cell counts were performed in whole retinas of 1-day (E), 3-days (F), 7-days (G). The cellular morphology and distribution of microglial cells in the superficial retina were assessed in 3-days (H) and 7-days (I). Activated microglial cells are indicated by white asterisks. Microglial cell counting was conducted in the superficial retina of 3-days (J) and 7-days (K). Data points for CD45⁺ cells were derived from four flat-mounted retinas, and data points for microglial cells were from five flat-mounted retinas. Data were presented as means $\pm$ s.e.m, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$, t-test. Scale bar = 50 μ m.

ligation of the unilateral common carotid artery (**Fig. 8H** [↗](#)). RNA-seq analysis of retinas subjected to 60 minutes of ischemia followed by 7-days of reperfusion in the HIOP model revealed enrichment of immune-related pathways, specifically highlighting increased activity in leukocyte-mediated immunity and regulation of immune effector processes (**Fig. 8B** [↗](#)). This finding was consistent with the results of GO analysis, which revealed associations of the overlapping DEGs between the HIOP and UPOAO models with toll-like receptor binding, T-cell receptor binding, and immune-related functions (**Fig. 8C, D** [↗](#)). To further explore these DEGs, we explored the remaining DEGs in the HIOP model in addition to the overlapping genes. GO analysis revealed that the remaining DEGs in the HIOP model were also enriched in immune responses, such as the T-cell receptor complex pathway, adaptive immune response, and B-cell-mediated immunity function (**Fig. 8E** [↗](#)). Remarkably, upon examining the remaining DEGs specific to the UPOAO model, GO analysis revealed DEGs distinctly related to lipid and steroid metabolic processes (**Fig. 8F, G** [↗](#)). This observation indicates that the UPOAO model, which is closely related to RAO, involves not only conventional immune responses but also unique regulation of lipid and steroid metabolism.

Similarly, the UCCAO model exhibited associations with negative regulation of B-cell proliferation, lymphocyte activation, and related processes (**Fig. 8I** [↗](#)). Additionally, the hub genes identified through protein–protein interaction (PPI) analysis are presented (**Fig. 8J** [↗](#)). However, the UCCAO model displayed minimal overlap with the UPOAO model, with only 15 overlapping DEGs identified in the Venn diagram analysis (**Fig. 8K** [↗](#)). Further analysis of the UCCAO DEGs revealed two genes (Mettl14 and YTHDF2) related to m6A modifications (**Fig. 8L** [↗](#)). Although arterial blood flow is occluded in both the UPOAO and UCCAO models, our results revealed that the pathophysiological processes of the two models were particularly different after 7-days of reperfusion.

4 Discussion

This study aimed to develop a more appropriate mouse model for investigating the ischemia-reperfusion process in RAO and unravelling the underlying pathophysiological mechanisms. We combined silicone wire embolization with carotid artery ligation to effectively block the blood supply to the retinal artery, closely mimicking the characteristics of the acute interruption of blood supply in RAO patients [\[31\]](#) [↗](#). In the UPOAO model, a 60-minute ischemia period was sufficient to simulate the injury of major retinal neural cells, such as RGCs, BCs, HCs, and cholinergic amacrine cells, resulting in visual impairment. Moreover, histologic examination demonstrated thinning of the inner layer of the retina, especially the GCL. Time course transcriptome analysis revealed various pathophysiological processes related to immune cell migration, oxidative stress, and immune inflammation during non-reperfusion and reperfusion periods. The resident microglia within the retina and peripheral leukocytes that access the retina were markedly increased during reperfusion periods. Additionally, we compared the transcriptomic signatures of the UPOAO model with those of two commonly used ischemia-reperfusion mouse models (HIOP and UCCAO), highlighting the potential relevance of the UPOAO model to ocular vascular occlusive diseases.

The UPOAO mouse model effectively simulates the characteristic features of RAO observed in patients. RAO patients often exhibit distinctive patterns in dark-adapted ERGs, where b-waves are reduced while a-waves remain stable [\[32–34\]](#) [↗](#). In our study, we observed almost the same phenomenon in ERG amplitude after 60 minutes of ischemia in the UPOAO model (**Table S2** [↗](#)). This observation was further confirmed by immunofluorescence staining of bipolar cells and photoreceptor cells. Importantly, a previous study linked the thinning of the inner retina with alterations in ERG in patients with RAO [\[35\]](#) [↗](#). Our results showed that ischemic damage predominantly affected the inner retinal layers, while the outer layer was almost unaffected (**Table S2** [↗](#)). This distinction is attributed to the reliance of the inner layers on the retinal artery

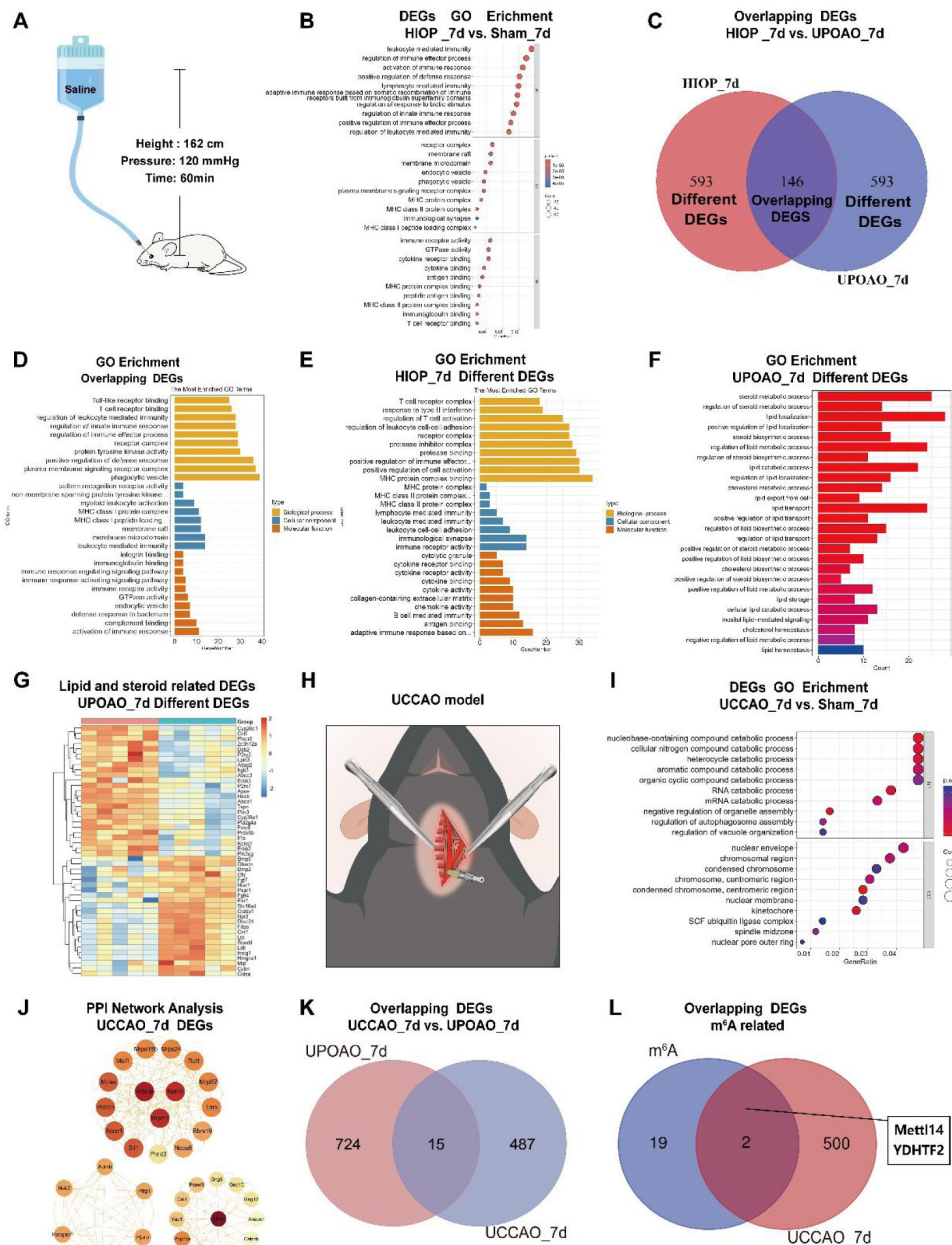


Fig. 8

Transcriptomic Results Comparison between UPOAO, HIOP, and UCCAO Models.

RNA-seq comparison between UPOAO and extravascular occlusion models: HIOP and UCCAO. (A) Schematic illustration of the HIOP model. (B) GO analysis of differential expression genes (DEGs) in the HIOP model at 7-days perfusion. (C) Venn diagram indicating the overlapping DEGs (146 genes) between HIOP and UPOAO models, as well as the remaining DEGs specific to each group (593 genes in each group). (D) GO analysis of the overlapping DEGs between HIOP and UPOAO models. (E) GO analysis of the remaining DEGs in the HIOP model at 7-days perfusion, excluding the overlapping DEGs. (F) GO analysis of the remaining DEGs, excluding the overlapping DEGs. (G) Heatmap showing the inter-sample distribution of lipid and steroid-related DEGs from the analysis of the remaining DEGs in the UPOAO model. (H) Schematic illustration of the UCCAO model. (I) GO analysis of DEGs in the UCCAO model. (J) Hub genes identified through protein-protein interaction (PPI) network analysis of the DEGs in the UCCAO model. (K) Venn diagram indicating the overlapping DEGs between UCCAO and UPOAO models (15 genes). (L) Venn diagram indicating the presence of two overlapping DEGs between UCCAO DEGs and m6A-related genes.

for blood supply, whereas the outer layers are supplied by the choroid. Additionally, histological examination revealed increased retinal NFL and GCL thickness and decreased IPL thickness after 3-days of reperfusion, indicating that ischemia-induced oedema had not completely subsided.

This study comprehensively explored the transcriptomic signature after ischemia and revealed that resident and peripheral immune cells may play major roles in pathological processes. In the non-reperfusion group, the transcriptional changes were primarily involved in the migration of immune cells such as leukocytes and neutrophils from blood vessels into ischemic retinal tissue. This process is closely linked to injury during ischemia and reperfusion, where leukocytes and neutrophils infiltrate neural tissue through the vascular endothelium [36, 37]. Our results revealed an increase in CD45⁺ leukocytes in the retina and retinal microvasculature, suggesting that these leukocytes were recruited from the bloodstream to the damaged retinal tissue following reperfusion. During leukocyte access to the retina, chemokines, chemokine receptors, adhesion molecules and cytoskeletal components play important roles in regulating endothelial permeability and facilitating leukocyte adhesion to endothelial cells [38, 39]. Our findings of related gene expression are consistent with this mechanism.

During the 3-days reperfusion period, we observed that an increased number of DEGs were significantly involved in the critical pathological response to oxidative stress in the context of IR tissues. The generation of reactive oxygen species (ROS) in mitochondria and subsequent oxidative stress are widely recognized as major causes of retinal cell damage induced by IR [40, 41]. This finding suggested that mitochondrial function, which is involved in oxidative stress processes, may play an important role in the pathophysiology of IRI during the early reperfusion stage. Oxidative stress can influence the permeability of the inner blood retina barrier (iBRB), allowing leukocytes to access retinal tissues [42]. The infiltration and activation of immune cells are recognized as the underlying causes of many retinal diseases, including ischemic ophthalmia [42, 43]. In the UPOAO model, the increase in CD45⁺ leukocytes and Iba1⁺ microglia in the retina after 3-days of reperfusion supports this viewpoint.

In the later stage of reperfusion (7-days), the DEGs were mainly enriched in immune regulation and inflammation. With more upregulated DEGs than downregulated ones, we hypothesize that the activation of the immune inflammatory response contributes to further IRI in retinal tissue. Previous studies have shown that the retina can elicit immunological responses during ischemia-reperfusion injury and that immune inflammation is an important phenomenon in the progression of this injury [30]. The excessive ROS generated by mitochondria in RGCs during the activation of inflammatory responses can damage cell structure and visual function [44]. Microglia, considered the primary resident immune cells, contribute to inflammatory responses and consequent neural damage [45]. Infiltrating leukocytes can activate resident microglia, forming a feedback loop that exacerbates inflammation [46]. Complex interactions between immune cells and retinal neurons after RIRI have been reported [47]. Interestingly, in our UPOAO model, the death of major retinal neural cells, such as RGCs, BCs, and HCs, was correlated with an increase in infiltrating leukocytes and resident microglia. Therefore, we considered that the immune response and neuroinflammation observed after 7-days of reperfusion may result from the cumulative effects of acute oxidative stress and the responses of resident and peripheral immune cells (**Fig. 9**, **Table S2**). In summary, we described the pathological processes at different time points and highlighted the important role of resident and peripheral immune cell responses in the UPOAO model. This finding offers valuable insights for subsequent screening of pathogenic genes and potential immunotherapy approaches.

Compared with the HIOP and UCCAO models, the UPOAO model is a more appropriate choice for studying retinal IRI in RAO. The HIOP model, widely used for investigating the pathogenesis of APACG, exhibits retinal degeneration features similar to those of the UPOAO model during both reperfusion periods [48]. In our UPOAO model, histologic and immunohistochemical analysis demonstrated that 60 minutes of ischemia followed by 3-days or 7-days of reperfusion can cause

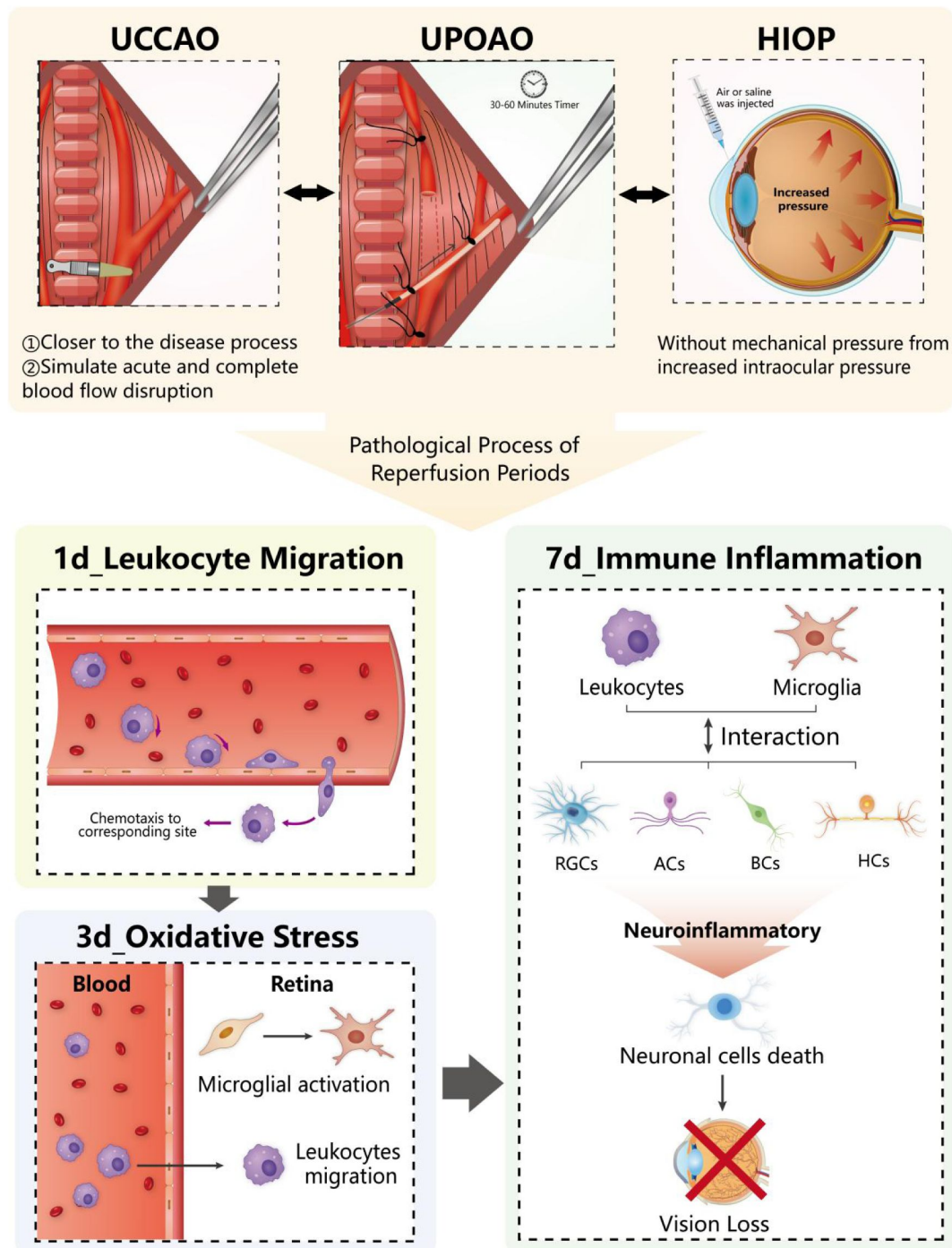


Fig. 9

The Characteristic Features of UPOAO Model during Ischemia-Reperfusion Periods.

irreversible damage to retinal structure and visual function. Specifically, we observed distinct alterations, such as thinning of the inner retina, substantial RGCs apoptosis, and decreased b-wave amplitude. Multiple studies on the HIOP model have reported consistent results [45, 49, 50]. However, some typical phenotypes in the HIOP model, such as thinning of the outer nuclear layer and the entire neuroretina and decreased a-wave amplitude, were not observed in the UPOAO model. Moreover, our transcriptome sequencing revealed a specific subset of DEGs unique to the UPOAO model, including *Apoe*, *Abca1*, *Ldlr*, *Cyp39a1*, and *Bmp6*. Notably, these DEGs were significantly enriched in pathways associated with lipid and steroid synthesis. Steroid and lipid metabolism homeostasis is disrupted under conditions of oxidative stress and inflammation [51]. In the UPOAO model, the observed lipid and steroid biological processes may result from the cumulative pathological responses triggered by IRI, as observed in the HIOP model. In the UCCAO model, RNA-seq revealed that DEGs were enriched in two main pathways: 1) epigenetic modification-related pathways, including nucleobase-containing compound catabolic process, RNA catabolic process, mRNA catabolic process and N6-methyladenosine (m6A) modification; and 2) cell death pathways, including regulation of autophagosome assembly, negative regulation of neuron death, and negative regulation of the neuronal apoptotic process. Epigenetic mechanisms may play a key role in the pathophysiology of ocular disease [52]. m6A, one of the most common RNA modifications, has been reported to regulate cell death processes, including apoptosis and autophagy, in the pathological process of IRI [53]. These results suggest that epigenetic mechanisms may significantly influence cell death during the 7-days reperfusion period in the UCCAO model. Lee et al. evaluated the characteristics of UCCAO without reperfusion using visual, histological and immunohistochemical approaches. Their results revealed delayed perfusion of the ipsilateral retina, thinning of the inner retinal layer 10 weeks after surgery, and a dramatic decrease in the amplitudes of b-waves on day 14 after UCCAO [15, 54]. This finding suggested that UCCAO primarily represents a model of retinal hypoperfusion injury and may not effectively reflect acute ischemic-induced structural and functional damage in RAO (Fig. 9). These results led us to consider the UPOAO model an effective experimental model for studying the pathological processes underlying acute ischemia and IRI.

Our research has certain limitations that should be acknowledged. First, our study focused on the changes that occurred within 60 minutes of ischemia and within the first 7-days of reperfusion in the UPOAO model. Further exploration is needed to understand the changes induced by longer reperfusion periods. Additionally, we proposed possible pathological mechanisms, including iBRB damage, oxidative stress and immune inflammation, mainly based on the enrichment results of DEGs at three reperfusion time points. More molecular experimental validation is required to substantiate these proposed mechanisms.

In summary, by thoroughly exploring the injury to major retinal neural cells, visual impairment and pathophysiological changes in the UPOAO model, we confirmed that this model can effectively simulate the acute ischemia-reperfusion processes in RAO. It serves as an ideal mouse model for investigating the underlying pathological mechanisms of ischemia and reperfusion. Furthermore, our UPOAO model holds great promise as a novel model for studies on pathogenic genes and potential therapeutic interventions for RAO.

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Availability of data and materials

All data are available from the corresponding author upon reasonable request.

Authors' contributions

Xiao X and Yang AH conceived the project and supervised, guided the research. Li Y, Wang YD and Feng JQ wrote the draft, Feng JQ, Wang CS, Xie H and Li ZY revised the manuscript. Wang YD and Li Y constructed the animal models and obtained the data. Wang YD performed the immunofluorescence staining, HE staining. Li Y and Feng JQ performed the ERG, analyzed the RNA-seq data, and created the visualizations. Wan YW and Wang YD performed the FFA and OCT imaging and Wan YW analyzed the data. Lv BY and Wang YD took pictures and videos of the UPOAO model, and Feng JQ made schematic diagrams of modeling. Li YM, Xie H, Chen T, Wang FX provided support for the technical methods and contributed to the discussion, and reviewed the manuscript. All authors critically contributed to the manuscript, and have read and approved the final manuscript.

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Ethics approval and consent to participate

All animal experiments were performed following the ethical guidelines outlined in the ARVO Statement for the Use of Animals in Ophthalmic and Vision Research, with the approval of the Experimental Animal Ethics Committee of Renmin Hospital of Wuhan University (approval number: WDRM-20220305A).

Supplementary data

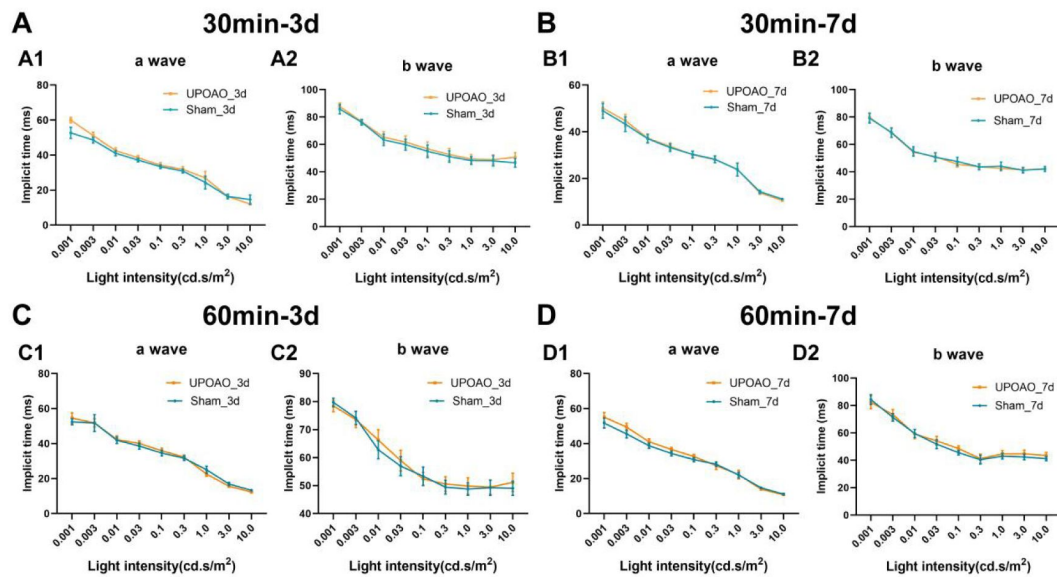


Fig.S1

Response Times of a-Waves and b-Waves in ERG at Different Light Intensities.

(A, B) Response times of a-waves and b-waves were measured under different light intensities at 3-days and 7-days post-UPOAO in the 30-minutes ischemia group. $n = 5$. (C, D) Response times of a-waves and b-waves were measured under different light intensities at 3-days and 7-days post-UPOAO in the 60-minute ischemia group. $n = 7$ at 3-days; $n = 8$ at 7-days. Data were presented as means $\pm$ s.e.m, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$, two-way ANOVA test.

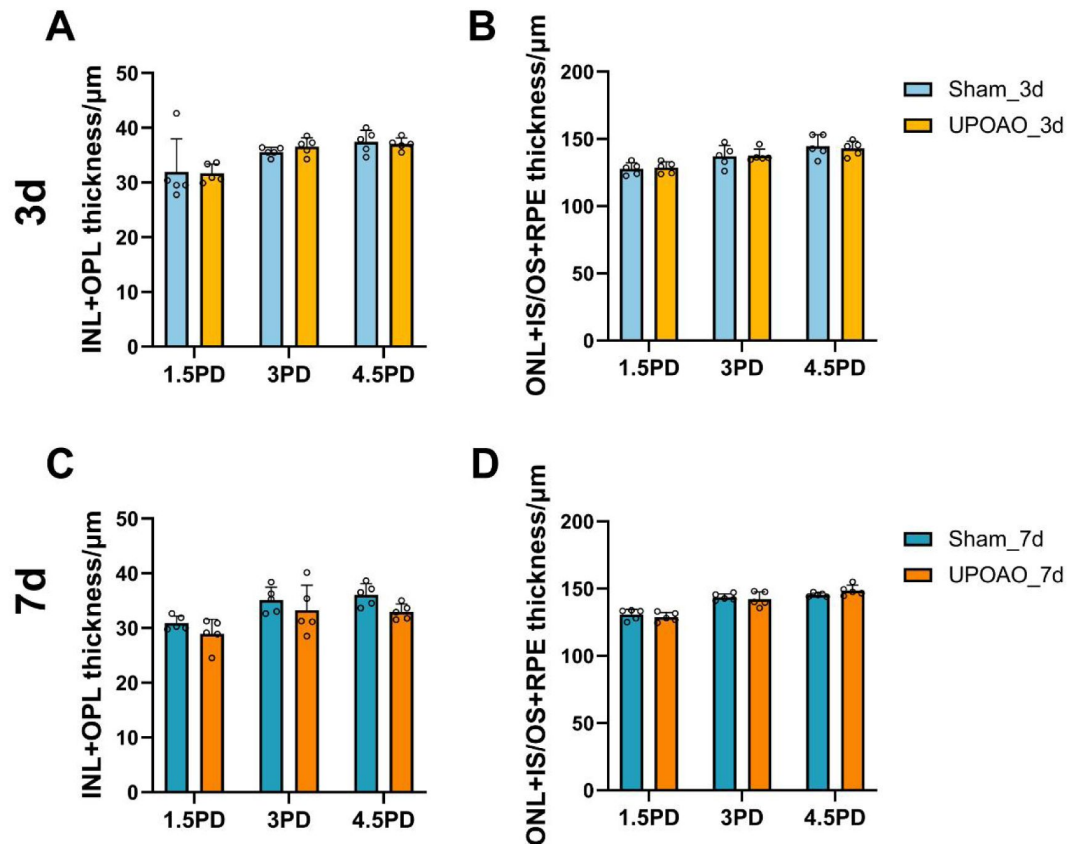


Fig.S2

Quantification of INL+OPL Thickness and ONL+IS/OS+RPE Thickness in OCT during 3-days and 7-days Reperfusion of UPOAO Animals.

The INL+OPL thickness and the ONL+IS/OS+RPE thickness were measured and compared at distances of 1.5 PD, 3.0 PD and 4.5 PD from the optic disc using OCT. (A and B) Quantification of INL+OPL layer thickness (A) and ONL+IS/OS+RPE layer thickness (B) at 3-days. $n = 5$. No significant differences were observed. (C and D) Quantification of INL+OPL layer thickness (C) and ONL+IS/OS+RPE layer thickness (D) at 7-days. $n = 5$. No significant differences were observed. Data were presented as means $\pm$ s.e.m, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$, two-way ANOVA test.

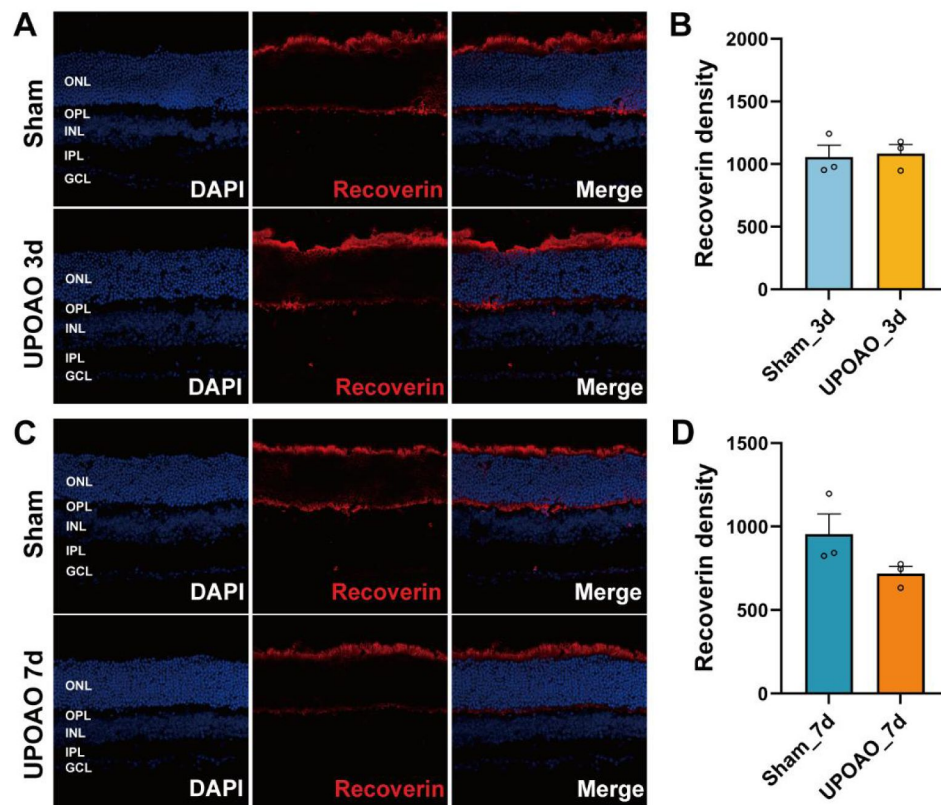


Fig.S3

Changes in Photoreceptor Cells in UPOAO.

(A) Representative images of mouse retina co-stained with DAPI and Recoverin at 3-days. (B) Representative images of mouse retina co-stained with DAPI and Recoverin at 7-days. $n = 3$. Data were presented as means $\pm$ s.e.m, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$, t-test. Scale bar = 50 μ m.

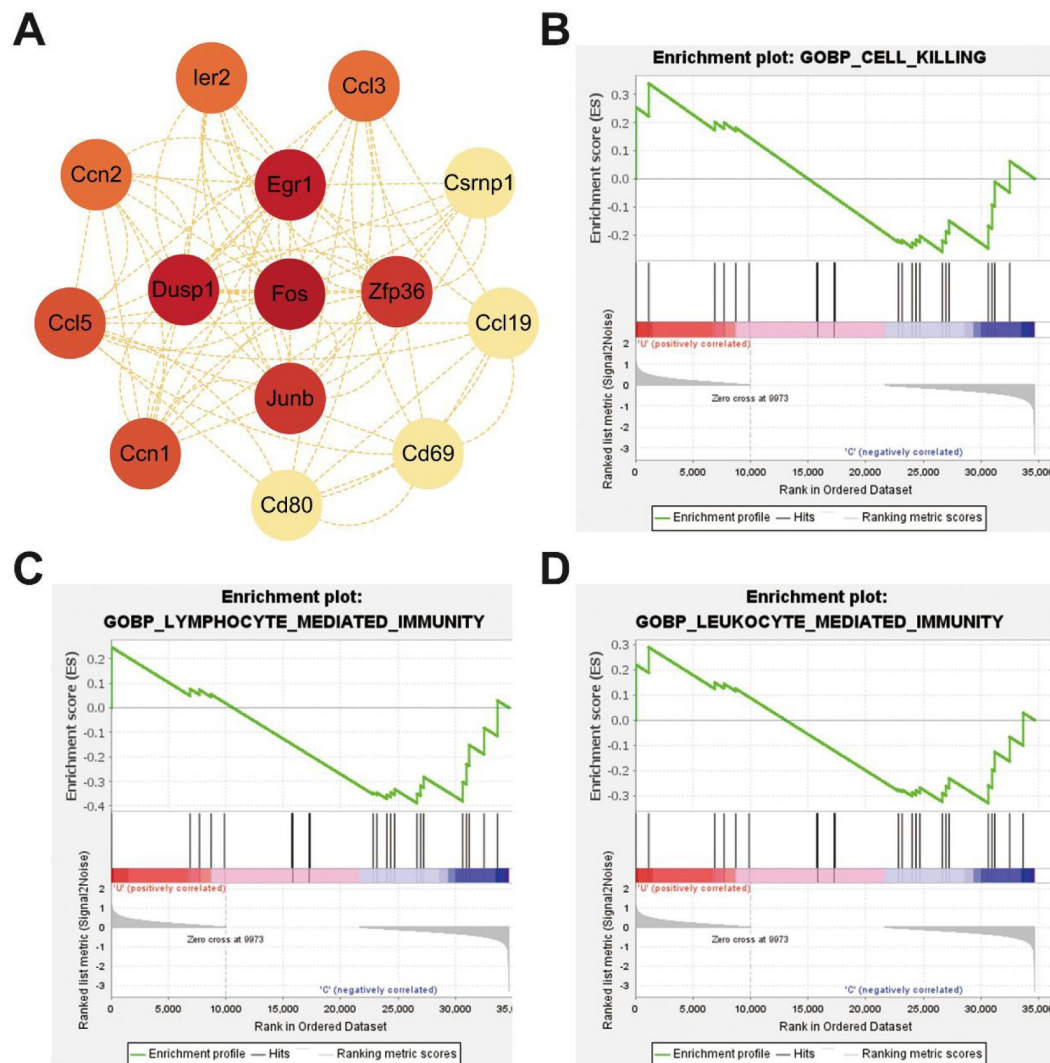


Fig.S4

Hub Genes in PPI Analysis and Gene Set Enrichment Analysis (GSEA) during the Non-Reperfusion Stage in UPOAO.

(A) Hub genes identified through PPI analysis of DEGs in the non-reperfusion group. (B-D) GSEA analysis of the pathway of the cell killing (B), lymphocyte mediated-immunity (C), and leukocyte mediated-immunity (D) in the non-reperfusion group.

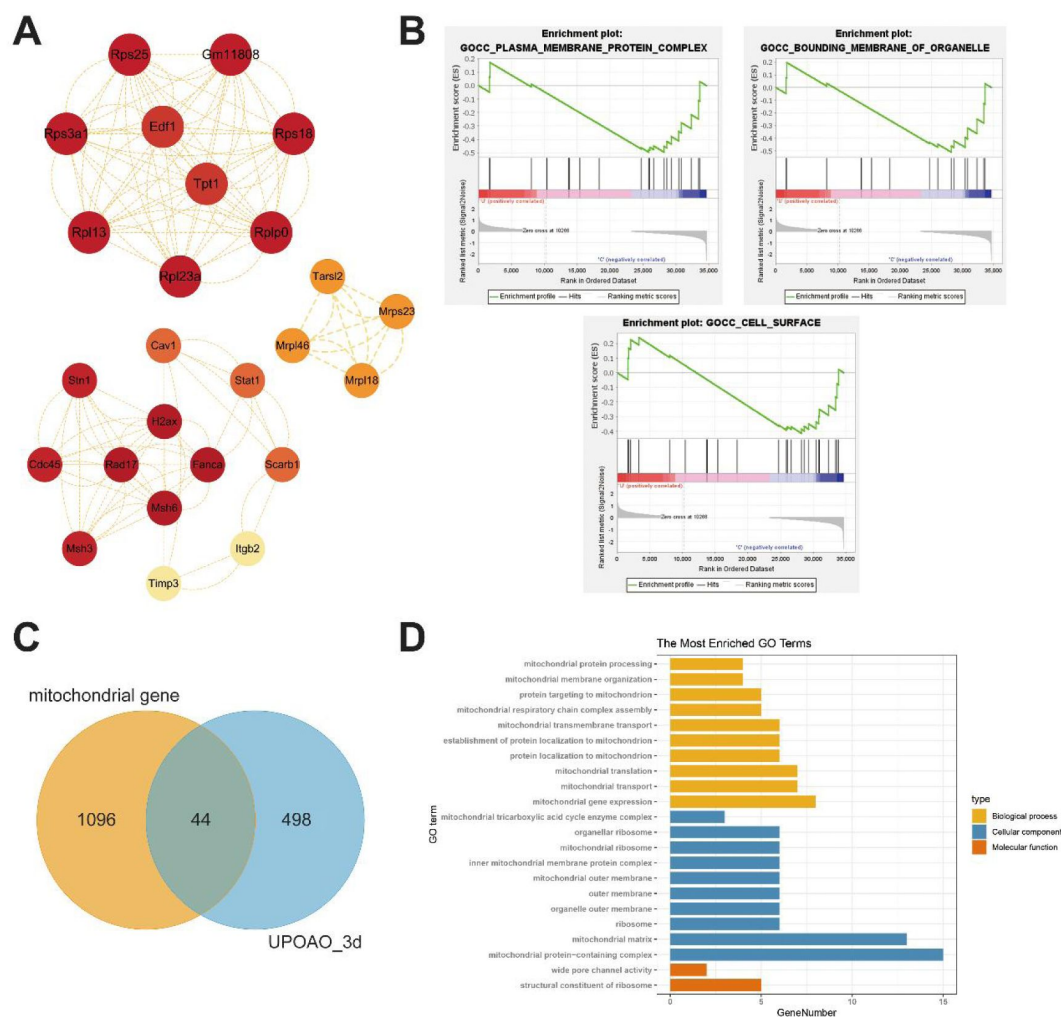


Fig.S5

Hub Genes, GSEA Analysis at 3-days Reperfusion in UPOAO, and Relation with Mitochondrial Genes.

(A) Hub genes identified through PPI analysis of DEGs in the 3-days reperfusion group. (B) GSEA analysis of the pathways related to plasma membrane protein complex (top left), bounding membrane of organelle (top right), and cell surface (bottom). (C) Venn diagram showing the overlapping genes between mitochondrial genes and DEGs at 3-days reperfusion (44 genes). (D) GO analysis of the 44 overlapping DEGs.

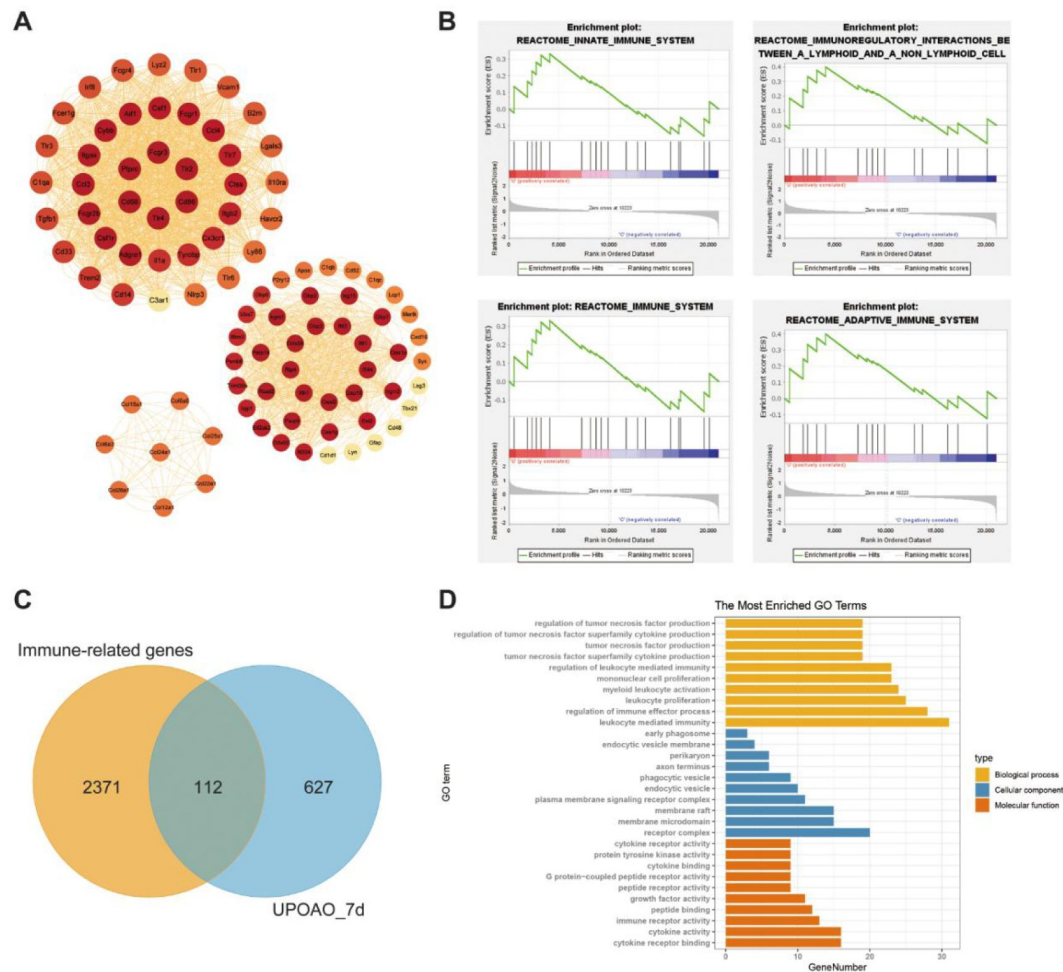


Fig.S6

Hub Genes, GSEA Analysis at 7-days Reperfusion in UPOAO, and Relation with Immune Genes.

(A) Hub genes identified through PPI analysis of DEGs in the 7-days reperfusion group. (B) GSEA analysis of the pathways related to the reactome innate immune system (top left), reactome immunoregulatory interactions between a lymphoid and a non-lymphoid cell (top right), reactome immune system (bottom left), and reactome adaptive immune system (bottom right). (C) Venn diagram showing the overlapping genes between immune genes and DEGs at 7-days reperfusion (112 genes). (D) GO analysis of the 112 overlapping DEGs.

Fig.S7

Upregulation of Immune Inflammation-Related Gene Expression in the 7-days Reperfusion Group.

The genes Abca1, Cd86, Cd48, Tlr4, Tlr6, and Tspo were significantly upregulated in the 7-days reperfusion group, while other immune-inflammatory and chemotaxis-related genes showed no significant differences. The data points were from the retina of four animals. Data were presented as means $\pm$ s.e.m, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$, Paired t-test.

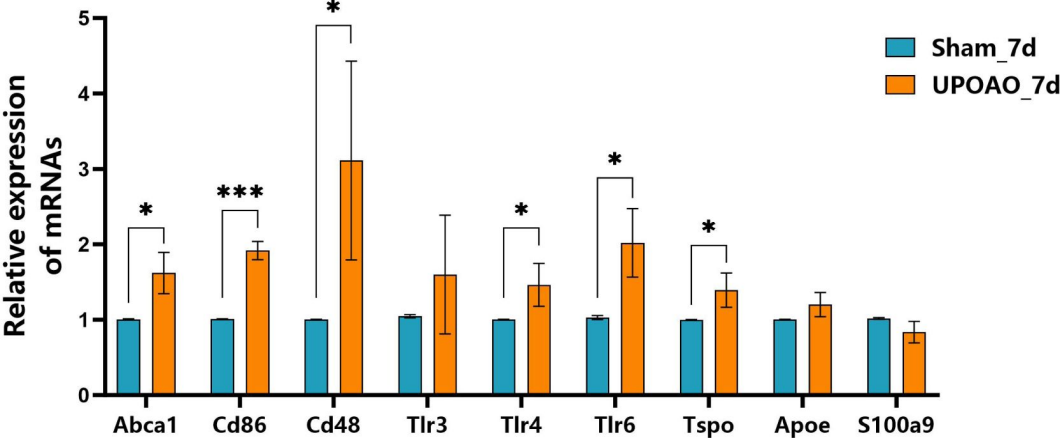
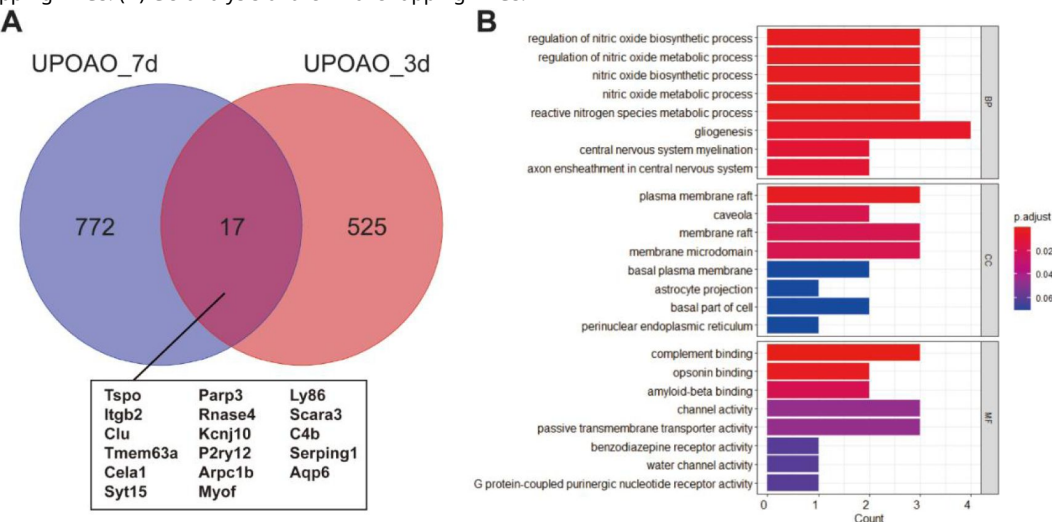


Fig.S8

Co-Expressed Genes during 3-days and 7-days Reperfusion.

(A) Venn diagram showing the overlap of DEGs between the 3-days and 7-days reperfusion groups, along with a list of the 17 overlapping DEGs. (B) GO analysis of the 17 overlapping DEGs.



Gene Name		Sequence (5' ->3')
Tlr3	Forward primer	CGATTCCCGACCAGCACTCT
	Reverse primer	CACATTTCTTTTTCAGCCTGTGAT
Tlr4	Forward primer	TCCCTGCATAGAGGTAGTTCC
	Reverse primer	TCAAGGGGTTGAAGCTCAGA
Cd48	Forward primer	GGAAGTGGATTTCAGGTCATTCA
	Reverse primer	ATGGTCCAAGTGGGTCCTTAT
Ldlr	Forward primer	CCAATCGACTCACGGGTTCA
	Reverse primer	ACAGTGTGCGACTTCTCTAGGC
Cd86	Forward primer	ATGGACCCCAGATGCACCA
	Reverse primer	TGTGCCCAAATAGTGCTCGT
Tspo	Forward primer	ACTGTATTCAGCCATGGGGTATG
	Reverse primer	CCAAGGGAACCATAGCGTCC
Abca1	Forward primer	GCAAGGCCGCACCATTATTT
	Reverse primer	TCTGAGAAACACTGTCCTCCTTTT
ApoE	Forward primer	CGGGTGCTCTGTTTTGGAGA
	Reverse primer	GGCTAGGCATCCTGTCAGC
S100a9	Forward primer	TGGCAACCTTTATGAAGAAAGAGA
	Reverse primer	GTGGGTTGTTCTCATGCAGC
β -actin	Forward primer	CACTGTCGAGTCGCGTCC
	Reverse primer	TCATCCATGGCGAACTGGTG

Table. S1

Primers Used in This Study.

Changes in each group		30min -3d	30min- 7d	60min-0d	60min-3d		60min-7d			
ERG	RGCs counts	-	-	/	↓		↓			
	Amplitudes of a-waves	-	-		-		- (0.001, 0.003, 0.01, 0.03, 0.1, 1.0 cd.s/m ²) ↓ (0.3, 2.0, 10.0 cd.s/m ²)			
	Response time of a-waves	-	-	/	-		-			
	Amplitudes of b-waves	-	-		↓		↓			
	Response time of b-waves	-	-		-		-			
	Amplitudes of OPs-waves		/		↓		↓			
OCT					1.5 PD	3 PD	4.5 PD	1.5 PD	3 PD	4.5 PD
	GCC thickness				-	-	-	↓	↓	↓
	INL+OPL thickness		/		-	-	-	-	-	-
	ONL+IS/OS+RP E thickness				-	-	-	-	-	-
	total retinal thickness				-	-	↓	↓	↓	↓
	RNFL+GCL thickness					↑			↓	
HE	IPL thickness					↓			↓	
	IINL thickness		/			-			↓	
	OPL thickness					-			-	
	ONL thickness					-			-	
	IS/OS+RPE thickness					-			-	
	Bipolar cells (PKCα)					-			↓	
Retina neural cells	Horizontal cells (Calbindin)					-			↓	
	Cholinergic amacrine cells (ChAT)		/			↓			↓	
	Photoreceptor cells (Recoverin)					-			-	
	Peripheral leukocyte infiltration (CD45) [#]		/			↑			↑	

Table. S2

Summary Table:

Time Course of all the Morphological, Functional, Cellular, and Transcriptome Changes in the UPOAO Model.

Changes in each group	30min-3d	30min-7d	60min-0d	60min-3d	60min-7d
Microglia counts (Iba1)		/		↑	↑
DEGs number			419 (up: 215; down: 204)	542 (up: 372; down: 170)	739 (up: 429; down: 310)
RNA-seq		/	Immune cells migration-related pathways	Oxidative stress-related pathways	Immune inflammation-related pathways
Enrichment					

-: no change;

↑ : increase;

↓ : decrease;

/: not involved in the article;

#: supplementary content: also increase after 60min-ischemia and 1d-reperfusion;

Table. S2 (continued)

Video 1. Blunt Dissection of Cervical Vessels. The neck was exposed, and the cervical arteries were separated using blunt dissection techniques.

Video 2. Insertion of Silicone Wire Embolus and Induction of Ischemia. The silicone wire embolus was inserted into the PPA, initiating the ischemic condition. PPA: the pterygopalatine artery.

Video 3. Extraction of Silicone Wire Embolus and Reperfusion. The silicone wire embolus was removed, allowing reperfusion to occur. The mouse's wound was sutured, and standard feeding procedures were followed.

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Reviewer #1 (Public Review):

Summary:

Wang, Y. et al. used a silicone wire embolus to definitively and acutely clot the pterygopalatine ophthalmic artery in addition to carotid artery ligation to completely block blood supply to the mouse inner retina, which mimic clinical acute retinal artery occlusion. A detailed characterization of this mouse model determined the time course of inner retina degeneration and associated functional deficits, which closely mimic human patients. Whole retina transcriptome profiling and comparison revealed distinct features associated with ischemia, reperfusion, and different model mechanisms. Interestingly and importantly, this team found a sequential event including reperfusion-induced leukocyte infiltration from blood vessels, residual microglial activation, and neuroinflammation that may lead to neuronal cell death.

Strengths:

Clear demonstration of the surgery procedure with informative illustrations, images, and superb surgical videos.

Two time points of ischemia and reperfusion were studied with convincing histological and in vivo data to demonstrate the time course of various changes in retinal neuronal cell survivals, ERG functions, and inner/outer retina thickness.

The transcriptome comparison among different retinal artery occlusion models provides informative evidence to differentiate these models.

The potential applications of the in vivo retinal ischemia-reperfusion model and relevant

readouts demonstrated by this study will certainly inspire further investigation of the dynamic morphological and functional changes of retinal neurons and glial cell responses during disease progression and before and after treatments.

Weaknesses:

It would be beneficial to the manuscript and the readers if the authors could improve the English of this manuscript by correcting obvious grammar errors, eliminating many of the acronyms that are not commonly used by the field, and providing a reason why this complicated but clever surgery procedure was designed and a summary table with time course of all the morphological, functional, cellular, and transcriptome changes associated with this model.

<https://doi.org/10.7554/eLife.98949.2.sa2>

Reviewer #2 (Public Review):

Summary:

The authors of this manuscript aim to develop a novel animal model to accurately simulate the retinal ischemic process in retinal artery occlusion (RAO). A unilateral pterygopalatine ophthalmic artery occlusion (UPOAO) mouse model was established using silicone wire embolization combined with carotid artery ligation. This manuscript provided data to show the changes of major classes of retinal neural cells and visual dysfunction following various durations of ischemia (30 minutes and 60 minutes) and reperfusion (3 days and 7 days) after UPOAO. Additionally, transcriptomics was utilized to investigate the transcriptional changes and elucidate changes in the pathophysiological process in the UPOAO model post-ischemia and reperfusion. Furthermore, the authors compared transcriptomic differences between the UPOAO model and other retinal ischemic-reperfusion models, including HIOP and UCCAO, and revealed unique pathological processes.

Strengths:

The UPOAO model represents a novel approach for studying retinal artery occlusion. The study is very comprehensive.

Weaknesses:

Originally, some statements were incorrect and confusing. However, the authors have made clarifications in the revised manuscript to avoid confusion.

<https://doi.org/10.7554/eLife.98949.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife assessment:

The manuscript establishes a sophisticated mouse model for acute retinal artery occlusion (RAO) by combining unilateral pterygopalatine ophthalmic artery occlusion (UPOAO) with a silicone wire embolus and carotid artery ligation, generating ischemia-reperfusion injury upon removal of the embolus. This clinically relevant model is useful for studying the cellular and molecular mechanisms of RAO. The data overall are solid,

presenting a novel tool for screening pathogenic genes and promoting further therapeutic research in RAO.

Thank you for recognizing the sophistication and clinical relevance of our mouse model for acute retinal artery occlusion. We are grateful for your supportive feedback.

Public reviews:

(1) Response to Reviewer #1:

Summary:

Wang, Y. et al. used a silicone wire embolus to definitively and acutely clot the pterygopalatine ophthalmic artery in addition to carotid artery ligation to completely block the blood supply to the mouse inner retina, which mimics clinical acute retinal artery occlusion. A detailed characterization of this mouse model determined the time course of inner retina degeneration and associated functional deficits, which closely mimic human patients. Whole retina transcriptome profiling and comparison revealed distinct features associated with ischemia, reperfusion, and different model mechanisms. Interestingly and importantly, this team found a sequential event including reperfusion-induced leukocyte infiltration from blood vessels, residual microglial activation, and neuroinflammation that may lead to neuronal cell death.

Strengths:

Clear demonstration of the surgery procedure with informative illustrations, images, and superb surgical videos.

Two-time points of ischemia and reperfusion were studied with convincing histological and in vivo data to demonstrate the time course of various changes in retinal neuronal cell survivals, ERG functions, and inner/outer retina thickness.

The transcriptome comparison among different retinal artery occlusion models provides informative evidence to differentiate these models.

The potential applications of the in vivo retinal ischemia-reperfusion model and relevant readouts demonstrated by this study will certainly inspire further investigation of the dynamic morphological and functional changes of retinal neurons and glial cell responses during disease progression and before and after treatments.

We sincerely appreciate your detailed and positive feedback. These evaluations are invaluable in highlighting the significance and impact of our work. Thank you for your thoughtful and supportive review.

Weaknesses:

It would be beneficial to the manuscript and the readers if the authors could improve the English of this manuscript by correcting obvious grammar errors, eliminating many of the acronyms that are not commonly used by the field, and providing a reason why this complicated but clever surgery procedure was designed and a summary table with the time course of all the morphological, functional, cellular, and transcriptome changes associated with this model.

Thank you for your thorough review of the manuscript. We sincerely apologize for any grammatical errors resulting from our English language proficiency and have taken the necessary steps to polish the article. Additionally, we have heeded your advice and reduced

the use of field-specific acronyms to enhance readability for both the manuscript and its readers.

Regarding the rationale behind the design of the UPOAO model, we have provided a description in Introduction section. Our group focuses on the research of pathogenesis and clinical treatment for RAO. The absence of an accurate mouse model simulating the retinal ischemic process has hampered progress in developing neuroprotective agents for RAO. To better simulate the retinal ischemic process and possible ischemia-reperfusion injury following RAO, we developed a novel vascular-associated mouse model called the unilateral pterygopalatine ophthalmic artery occlusion (UPOAO) model. We drew inspiration from the widely employed middle cerebral artery occlusion (MCAO) model, commonly used in cerebral ischemic injury research, which guided the development of the UPOAO model.

We appreciate your valuable suggestion regarding the inclusion of a summary table outlining the time course of morphological, functional, cellular, and transcriptome changes associated with this model. To address this, we intend to include a supplementary table at the end of the article (Table. S2 Summary Table), which will offer a comprehensive overview of the experimental results, thereby aiding in clarity and interpretation.

Once again, we thank you for your insightful comments and suggestions, which have greatly contributed to the improvement of our manuscript.

(2) Response to Reviewer #2:

Summary:

The authors of this manuscript aim to develop a novel animal model to accurately simulate the retinal ischemic process in retinal artery occlusion (RAO). A unilateral pterygopalatine ophthalmic artery occlusion (UPOAO) mouse model was established using silicone wire embolization combined with carotid artery ligation. This manuscript provided data to show the changes in major classes of retinal neural cells and visual dysfunction following various durations of ischemia (30 minutes and 60 minutes) and reperfusion (3 days and 7 days) after UPOAO. Additionally, transcriptomics was utilized to investigate the transcriptional changes and elucidate changes in the pathophysiological process in the UPOAO model post-ischemia and reperfusion. Furthermore, the authors compared transcriptomic differences between the UPOAO model and other retinal ischemic-reperfusion models, including HIOP and UCCAO, and revealed unique pathological processes.

Strengths:

The UPOAO model represents a novel approach to studying retinal artery occlusion. The study is very comprehensive.

We greatly appreciate your positive assessment of our work and are encouraged by your recognition of its significance.

Weaknesses:

Some statements are incorrect and confusing. It would be helpful to review and clarify these to ensure accuracy and improve readability.

We sincerely appreciate your meticulous review of the manuscript. Taking into account your valuable feedback, we will thoroughly address the inaccuracies identified in the revised version. Additionally, we will commit to polishing the article to ensure improved readability. We apologize for any confusion caused by these inaccuracies and genuinely thank you for bringing them to our attention.

Recommendations For The Authors:

Reviewer #1:

(1) Response to comment:

The conclusions of this paper are mostly well supported by clear images and convincing data analysis, but some aspects of image presentation and additional data analysis may be needed to strengthen the manuscript.

We sincerely appreciate your positive assessment of our work and your recognition of the clear images and convincing data analysis supporting our conclusions. Your constructive feedback on enhancing the clarity of our manuscript's image presentation and additional data analysis is highly valued. In response to your suggestions, we have taken steps to improve readability by removing or correcting uncommon acronyms from certain images. We have also conducted further data analysis to provide more comprehensive insights. Thank you for your guidance in improving the quality of our manuscript.

(2) Response to recommendation (1):

In Results 3.1 or in Method 2.2: please explain why this combination of silicone wire embolization and carotid artery ligation was chosen to replace previous models such as UCCAO? What are the advantages? And why the silicone wire embolus was inserted through ECA instead of inserting into CCA directly? The cleverly designed surgical procedure is very impressive but the reasoning behind it is not obvious and needs more explanation.

Thank you for your valuable feedback.

In the introduction, we briefly describe the rationale for developing the UPOAO model to simulate acute ischemia-reperfusion of retinal artery occlusion (RAO). Previous common retinal ischemia model had certain shortcomings. For example, in the HIOP model, which is often used for simulating glaucoma, the ischemic factor of interrupted retinal blood flow may be amplified due to the dual effects of IOP-induced mechanical stress [1, 2] and vascular ischemia due to normal saline perfusion in the anterior chamber. In the UCCAO model, recanalization is performed after ligation of the carotid blood vessels, and the retina communicates with the blood vessels in the brain, resulting in retinal hypoperfusion. The retina ischemia in UCCAO is a chronic process, for example, the retina became thinner at week 10 and week 15 [3], while RAO is an acute total retinal ischemic disease. Therefore, it is critically important to develop a simple mouse model that can simulate acute retinal ischemia and reperfusion injury in RAO patients.

Various models have been developed for ischemic stroke research, with the endoluminal suture model being the most employed method for middle cerebral artery occlusion (MCAO). In this model, filaments are introduced through either the external or internal carotid artery and advanced into the middle cerebral artery, causing temporary blood flow blockage for a specific duration. This method has been extensively employed in studies involving transient occlusion [4]. Among the MCAO models, the Koizumi method (occlusion from the common carotid artery (CCA) to the middle cerebral artery (MCA)) and the Longa method (occlusion from the external carotid artery (ECA) to the MCA) are frequently used. Among these two methods, the Longa method is more widely utilized in research studies. The Longa method has a much lower mortality rate post-surgery (26%) than that of the Koizumi (44%) [5]. The MCAO model induces substantial infarct areas and significantly contributes to advancements in stroke research, including investigations into blood-brain barrier disruption and inflammatory responses to ischemia.

RAO is considered a form of ocular stroke. Inspired by the MCAO model, we have employed a silicone wire embolus to induce acute interruption of blood flow to the retina. This approach enables the investigation of pathophysiological processes associated with RAO, providing valuable insights into the understanding of this condition. We have clarified these points in the revised manuscript (line 129).

The reasoning behind inserting the silicone wire embolus through the ECA instead of directly into the CCA is twofold:

- (1) Convenience and avoidance of heavy bleeding and mortality. Inserting the silicone wire embolus requires creating an opening in the artery, which then needs to be ligated at both ends after the silicone wire embolus is removed to prevent excessive bleeding. The ECA's ability to form a straight line with the ICA after folding makes it more convenient for the entry and removal of the silicone wire embolus. This procedure is more convenient to perform on the ECA. The blood flow to the CCA can be restored after the plug is removed from ECA, ensuring that the blood supply to the brain through the CCA is not affected.
- (2) Preservation of reperfusion process. If the silicone wire embolus were inserted directly into the CCA, the ends of the CCA opening would need to be ligated after the silicone wire embolus is removed. This would result in a lack of reperfusion process after retinal ischemia. To enable the reperfusion process, the decision was made to open the ECA instead.

We have clarified these points in the revised manuscript to better explain the rationale behind our methodology (line 139). Thank you for prompting this important clarification, which we believe will enhance the understanding of our readers.

(3) Response to recommendation (2):

Did the UPOPA actually block OA, including both the retinal (CRA) and choroidal (SPCA and LPCA) blood supply? If so, why does it seem only the inner retina was affected but not the outer retina?

Thank you for your question. We agree with you that the UPOAO model blocks OA, which includes retinal and choroidal vessels. Our experimental results primarily indicate damage to the inner retinal layer within 7 days of reperfusion. For example, OCT and HE staining showed significant thinning of the inner retina after 60 minutes of ischemia followed by 7 days of reperfusion (Figure 4). At the same time, the b-wave amplitudes were decreases, usually indicating damage to the inner layer of the retina. However, the outer retina was seemed not affected by 60 minutes of ischemia based on the results of OCT, HE and immunofluorescence.

Inner layer of the retina was known to show the highest sensitivity to hypoxic challenges [6], whereas the outer retinal layer was more resistant to hypoxic stress [7]. The possible reason for these results was that the outer layer like photoreceptors is more tolerant against ischemia than inner layer of the retina. Previous studies of retinal ischemia-reperfusion models supported this assumption. In the UCCAO model, the b-wave was more affected than the a-wave. Decreases in the amplitudes of OPs, scotopic b-wave, and photopic b-wave were consistently observed on week 4 after UCCAO, while the amplitude of scotopic a-wave did not dramatically change [8]. Prolonged ischemia, such as permanent ischemia, led to photoreceptor cell degradation, as seen in Stevens et al.'s report of photoreceptors loss 3 months after permanent ligation of both common carotid arteries in bilateral common carotid artery occlusion (BCCAO) [9]. In the HIOP model, the GCL and INL reacted sensitively to ischemic processes. A significant thinning of the GCL as early as 6 hours after 60 minutes of ischemia [10]. Horizontal cells and photoreceptors remained mostly unaffected, while most RGCs and several amacrine cell subtypes disappear [11, 12].

Our study revealed the changes that occurred within 60 minutes of ischemia and the first 7 days of reperfusion in the UPOAO model. One possibility was that the ischemia duration in our model was not long enough to affect the outer retinal cells. Furthermore, the observation time point for reperfusion was not long enough to see the structure damage and visual dysfunctions in the outer retinal layer. As we have explained in the manuscript, further exploration is needed to understand changes induced by longer ischemia duration and reperfusion periods. Revealing the damage to retinal structure and function during longer ischemia time will be an emphasis direction for our further research.

(4) Response to recommendation (3):

Better to only use well-accepted acronyms and remove those that are rarely seen in other publications, such as IMRL, MRL, HIOP, TRT, etc.

Thank you for your valuable feedback. In our manuscript, we utilized the Spectralis HRA+OCT device (Heidelberg) to capture the retinal images. However, the resulting image layering did not adequately distinguish each retinal layer clearly. To address this limitation, we referred to a clinical OCT stratification approach in RVO and divided the retina into the inner, middle, and outer layers [16]. We acknowledge that this hierarchical description is not commonly used and have therefore followed your recommendation to remove these rare acronyms and instead employ the layer structure abbreviation along with the plus sign. The methods and results have been revised accordingly (line 213, line 368, Figure 4 and Figure S2).

In addition, for the HIOP model, it is also known as the IR or RIRI model [17-19], and the pathophysiological process of retinal ischemia-reperfusion injury (IRI) is usually used to represent this type of anterior chamber perfusion model. To avoid confusion between the pathophysiological process of ischemia-reperfusion studied in this paper and the common model of high intraocular pressure, we have consistently referred to it as the HIOP model, an abbreviation that is cited in many references [20-22].

Thanks again for the suggestion. We apologize for any confusion caused by the use of abbreviations and have made the necessary corrections in the manuscript. We have also strengthened the details of OCT layering in the images to enhance readability for our audience.

(5) Response to recommendation (4):

Figure 3F, G: What do the OP changes mean? What retina cell dysfunction leads to OP changes? Is there RGC-relevant visual function readout to correlate with RGC death?

Oscillatory potentials (OPs) are important components of the electroretinogram (ERG). While the precise origin of OPs remains unclear, they are generally believed to be generated from the inner retinal layer, specifically involving bipolar cells, amacrine cells and ganglion cells [23]. OPs are sensitive indicators of retinal ischemic effects and can detect dysfunction before alterations in the b-waves occur [24-26] (We have added these statements at line 358). In this research, the reduction of OPs indicated dysfunction in the inner retinal layer and retinal ischemia.

The function of RGCs can be non-invasively assessed by using various ERG technique that emphasize the activity of inner retina neurons, including OPs of multifocal ERG (mfERG), photopic negative response (PhNR) in mfERG, pattern electroretinogram (PERG), negative Scotopic Threshold Response (nSTR) [27]. Among these indicators, the PERG appears to be more specifically related to the presence of functional RGCs. However, the complexity of electrophysiological sources and species-specific differences in RGCs characteristics should

also be considered. In addition, visual evoked potentials (VEP) can assess the function of visual signaling in the whole visual pathway from RGC axons to the visual cortex of the brain [28, 29]. Unfortunately, due to the unavailability of specific equipment required for evaluating RGCs function, we encountered limitations in conducting a comprehensive assessment in this study. This limitation emphasizes the importance of future studies incorporating RGCs evaluation to provide a more comprehensive understanding of visual pathway functionality and its implications, considering indicators such as PERG and PhNR.

Thank you for your careful review and insightful questions.

(6) Response to recommendation (5):

Figure 4B: RNFL/GCL/IPL normally called GCC (ganglion cell complex).

We appreciate your helpful recommendation regarding the abbreviation GCC (ganglion cell complex) for the combination of RNFL, GCL, and IPL. We have updated this terminology in the revised manuscript (line 213 and Figure 4).

(7) Response to recommendation (6):

Figure 4 A-F: Normally a circular OCT image surrounding the optic nerve head is preferred to measure retina thickness. If in these figures, all the OCT images are from the same location, it may be acceptable, but need to provide imaging details on how these OCT planes are selected and what has been done to make sure the same locations were selected for comparison.

We agree with your comment on OCT imaging that the retina is usually captured OCT images surrounding the optic nerve head. In this study, our goal was to assess both the thickness of the peripheral retina and the retina near the optic nerve head. To achieve this, we considered the optic nerve head as the apex of the selected field of view (left upper region of panel A in Figure 4). For each mouse, we obtained OCT images of the superior nasal (SN), superior temporal (ST), inferior nasal (IN), and inferior temporal (IT) fields of the optic nerve. We then averaged the thicknesses from these four fields. In each field, we measured and statistically evaluated the retinal thickness at distances of 1.5, 3, and 4.5 papillae diameters (PD) from the optic nerve head.

This approach allowed us to ensure that the same locations were selected for comparison and provided a comprehensive assessment of retinal thickness across different regions. We have detailed this methodology in the revised manuscript to clarify the imaging process and the consistency of the selected locations.

Thank you for your insightful feedback.

Reviewer #2:

Addressing the following concerns is necessary to improve the manuscript.

(1) Response to recommendation (1):

The manuscript contains many grammatical errors and should be carefully reviewed for corrections. For example: In the title, "Silicone Wire Embolization-induced Acute Retinal Artery Ischemia and Reperfusion Model in Mouse: Gene Expression Provide Insight into Pathological Processes". It should be "Provides" instead of "Provide". In the Abstract, "The resident microglia within the retina and peripheral leukocytes which access to the retina were pronounced increased on reperfusion periods." It should be "pronouncedly" or "markedly" instead of "pronounced".

Thank you for your careful reading and pointing out the grammatical errors in the manuscript. We apologize for these mistakes and have since revised and polished the article with the assistance of native English speakers. Ensuring accurate and clear language usage in scientific writing is crucial, and we appreciate your help in improving the quality of our manuscript. Thank you for bringing these errors to our attention.

(2) Response to recommendation (2):

Video 2: the video content from "30s-47s" and "50s-67s" is repeatedly shown.

Thank you for your careful review of the video. In the process of preparing the external carotid artery for silicone wire embolus insertion, we first ligated the distal end with a square knot and then tied a loose knot at the proximal end. In the video content from "30s-47s" and "50s-67s", we are tying a square knot. We apologize for any confusion caused by these repeated video clips.

(3) Response to recommendation (3):

Figure 1: The ConA staining (H-I) and FFA (J-K) were performed before the removal of silicone wire embolus. It would be beneficial to clarify this in the figure legend too. Additionally, the label 'Post. Sup. Alveolar art.: Posterior superior alveolar artery' is not present in Figure 1L."

Thank you for your thorough review of the manuscript and the valuable suggestions regarding Figure 1. We have updated the figure legend of Figure 1 to clarify that ConA staining (H-I) and FFA (J-K) were performed before the removal of the silicone wire embolus (line 868 and line 873). Additionally, we have included the label 'Post. Sup. Alveolar art' in Figure 1L as you pointed out. We appreciate your careful attention to detail, and we have ensured that these omissions have been rectified in the revised version of the manuscript.

(4) Response to recommendation (4):

Figure 2: only representative images of RGCs at the peripheral retina were shown. It is not clear if only RGCs in the peripheral retina were quantified. Is there RGC loss in the central and middle retina in the UPOAO model as well? How many fields of RGCs were quantified for each retina?

Thank you for your meticulous review of the manuscript. The quantification method of RGCs is described in detail as follows:

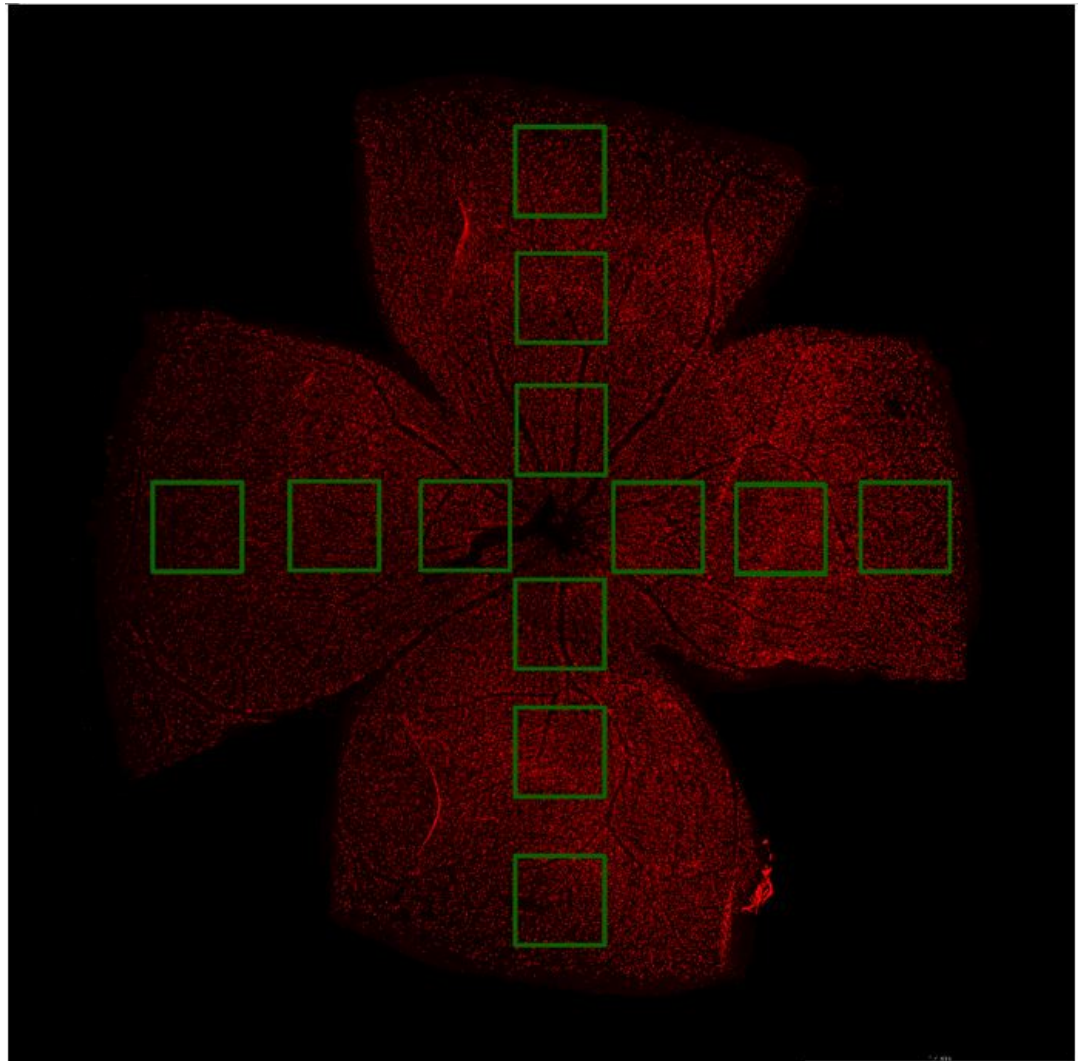
Four radial incisions were made in the retina and flattened on a glass slide to create a "four-leaf clover" shape. Retina was photographed using a fluorescence microscope (BX63, Olympus, Japan). We captured images from three different regions of each retinal quadrant: 0.1 mm-0.5 mm (central region, field numbers: 1, 4, 7, 10), 0.9 mm-1.3 mm (middle region, field numbers: 2, 5, 8, 11), and 1.7 mm-2.1 mm (peripheral region, field numbers: 3, 6, 9, 12) from the optic nerve head, respectively, as shown in Author response image 1.

Of these, the peripheral field changes were the most noticeable, so we used the Leica SP8 confocal microscope (20X) to capture peripheral field RGCs as a demonstration (Figure 2A, C, E, G). RGC counts of twelve fields of each retina were quantified and the average density of RGCs in twelve fields per retina was shown in Figure 2B, D, F, K. RGC counts in the central (field number: 1, 4, 7, 10), middle (field number: 2, 5, 8, 11), and peripheral (field number: 3, 6, 9, 12) visual fields were shown in Author response table 1-4. We have included this detailed methodology in the revised manuscript to clarify the quantification process and to address

the presence of RGCs loss in both the central and middle retina in the UPOAO model. Thank you for pointing out the need for this clarification.

Author response image 1.

Schematic diagram of field selection. Scale bar=1.4 mm. Each retinal petal has three distinct visual fields (the area circled by the green line) that radiate from the optic nerve head to the periphery, in that order, the central, middle, and peripheral visual fields.



Author response table 1.

RGCs counts in each field of each retina (30-minute ischemia and 3-day reperfusion)

group	Sham			UPOAO		
field/sample	1	2	3	1	2	3
1	529	707	763	559	609	552
2	582	723	700	553	522	476
3	328	404	425	284	284	265
4	593	643	671	509	643	550
5	613	629	760	553	646	568
6	611	363	751	590	393	577
7	539	697	696	501	600	554
8	595	626	451	641	613	627
9	585	415	288	492	593	453
10	581	656	647	499	547	556
11	685	707	801	607	564	673
12	438	691	458	526	524	590
average	557	605	618	526	545	537
density	3333	3623	3698	3151	3262	3214

Author response table 2.

RGCs counts in each field of each retina (30-minute ischemia and 7-day reperfusion)

group	Sham			UPOAO		
field/sample	1	2	3	1	2	3
1	802	775	777	696	491	651
2	612	640	812	764	477	519
3	328	349	588	400	222	102
4	768	740	732	729	523	616
5	802	780	887	879	618	636
6	525	476	742	713	364	713
7	724	621	790	741	589	738
8	720	716	513	705	442	742
9	562	489	304	416	112	173
10	795	758	739	598	518	774
11	727	812	831	643	462	722
12	387	680	661	367	486	303
average	646	653	698	638	442	557
density	3868	3910	4180	3818	2647	3338

Author response table 3.

RGCs counts in each field of each retina (60-minute ischemia and 3-day reperfusion)

group	Sham					UPOAO				
field/sa	1	2	3	4	5	1	2	3	4	5
mple										
1	812	668	797	736	750	429	505	725	792	710
2	731	619	824	857	652	644	251	778	710	755
3	524	568	612	470	280	297	338	321	459	434
4	786	793	794	764	684	476	628	650	654	691
5	799	727	859	755	685	570	533	673	721	675
6	657	503	569	619	398	405	509	470	684	523
7	729	681	666	713	637	684	484	772	711	704
8	731	775	758	583	779	663	614	805	564	636
9	602	536	432	670	662	391	329	470	499	184
10	695	676	687	738	557	456	576	663	683	691
11	598	714	736	754	670	564	478	700	791	610
12	487	669	589	695	388	421	669	573	612	112
average	679	661	694	696	595	500	493	633	657	560
density	4067	3957	4153	4169	3564	2994	2951	3792	3932	3356

Author response table 4.

RGCs counts in each field of each retina (60-minute ischemia and 7-day reperfusion)

group	Sham					UPOAO				
field/sample	1	2	3	4	5	1	2	3	4	5
1	690	689	681	768	523	271	362	129	282	164
2	695	517	690	832	671	269	354	154	99	154
3	451	307	505	669	607	138	169	86	19	123
4	643	664	618	709	604	287	323	140	260	184
5	699	683	670	721	540	278	301	124	346	146
6	621	561	635	654	358	204	111	79	178	120
7	687	604	690	769	629	295	433	169	315	126
8	490	730	752	724	617	350	314	84	257	25
9	269	606	573	731	491	91	62	9	175	16
10	681	741	658	714	499	346	362	151	359	129
11	814	836	546	674	665	285	379	152	198	93
12	632	625	357	324	625	67	231	34	140	22
average	614	630	615	691	569	240	283	109	219	109
density	3679	3774	3680	4136	3408	1438	1697	654	1311	650

(5) Response to recommendation (5):

Figure 3: The representative wave lines in panels A (60min_3d, 60min_7d) and F do not reflect the statistical analysis presented in panels D, E, and G, especially for the amplitudes of b waves and OPs.

Thank you for your careful review of the manuscript. We've added labels for a-waves, b-waves, and improved the presentation of OPs to make the details of the amplitude more visible (Figure 3). In the previous version, due to incorrect settings, we did not adjust the ordinate spacing when fitting curves of representative wave lines in four groups, resulting in the curves being compressed vertically to the same height. We have now adjusted the curves to be fitted under the same scale bar (shown in the bottom right corner of Figure. 3A). What's else, we removed the baseline wave of the OPs wave and adjusted the abscissa scale to highlight the N waves and P waves for easy reading (Figure 3F).

(6) Response to recommendation (6):

There are two different Supplementary Figure 1 and no Supplementary Figure 3, resulting in misaligned references to Supplementary Figures 1, 2, and 3 in the text.

Thank you for your careful review of the manuscript. We have reviewed the manuscript again and identified errors in uploading the supplementary figures, which resulted in duplicate Supplementary Figure 1 and the absence of Supplementary Figure 3. We have

corrected these issues and realigned the references to Supplementary Figures 1, 2, and 3 in the text to ensure consistency. We appreciate your attention to detail and your reminder to address this issue.

(7) Response to recommendation (7):

There is confusion about the definition of ORL (outer retina layer). In Lines 208-209, ORL was defined as the combined thickness of the rest to the retinal pigment epithelium (RPE). It seems the ONL is included in ORL. But in lines 358-359, 907-908, "the ORL encompassed the region from the inner segment/outer segment (IS/OS) to the RPE". Please make the definition consistent. In addition, it is hard to distinguish the regions marked by the green lines in Fig. 4A (sham image) after Line 902.

Thank you for your careful review of the manuscript. We have addressed the confusion regarding the definition of the outer retinal layer (ORL). The Heidelberg OCT device does not distinguish the layers of the mouse retina well, so we divided it into three broader layers:

- (1) Ganglion Cell Complex (GCC) layer, which encompasses RNFL+GCL+IPL.
- (2) Middle Retinal Layer, which includes INL+OPL.
- (3) Outer Retinal Layer (ORL), which includes ONL+IS/OS+RPE.

We apologize for the inconsistency and have revised the errors in the manuscript and figure legends accordingly. Additionally, we have removed rare domain-specific acronyms and replaced them with more commonly understood abbreviations, as suggested, to avoid confusion.

Furthermore, we have enlarged parts of the OCT images to better display the layers, hoping to meet the readers' requirements and improve clarity. Thank you for your valuable feedback.

(8) Response to recommendation (8):

Figure 4 (Panels H-J, L-M) incorporated with the text (Line 902) differs from the high-resolution version of Figure 4 included later in the manuscript. In Figure 4 (Panels H-J, L-M) merged with the text (Line 902), the quantification of the IPL and INL thickness is incorrect, and the scale bar is inaccurate. However in the high-resolution version of Figure 4 provided later, the thickness of the RNFL+GCL is incorrect.

Thank you for your careful review of the manuscript. The quantification of the IPL and INL thickness in Figure 4 (Panels H-J, L-M) incorporated with the text has been revised to ensure accurate measurements and scale bars (Figure 4 and line 924). The high-resolution version of Figure 4 provided later has been updated to correct the thickness measurements of the RNFL+GCL. We have ensured that the ordinate in the high-resolution version of Figure 4 now correctly represents length units, consistent with the equal proportional conversion used in the integrated text figures.

Thank you for your valuable feedback and for pointing out these errors. We have made the necessary corrections to align the figures accurately with the manuscript.

(9) Response to recommendation (9):

Line 384-386: the statement "Notably, a-waves in ERG and the thickness of the outer retinal layers in both OCT and HE remained unchanged." is not accurate, since a-waves in ERG is not changed in 3 days but changed in 7 days, and the thickness of the outer retinal layers in HE is either not measured or not shown in Figure 4.

Thank you for your careful review of the manuscript. We apologize for this error and have revised it.

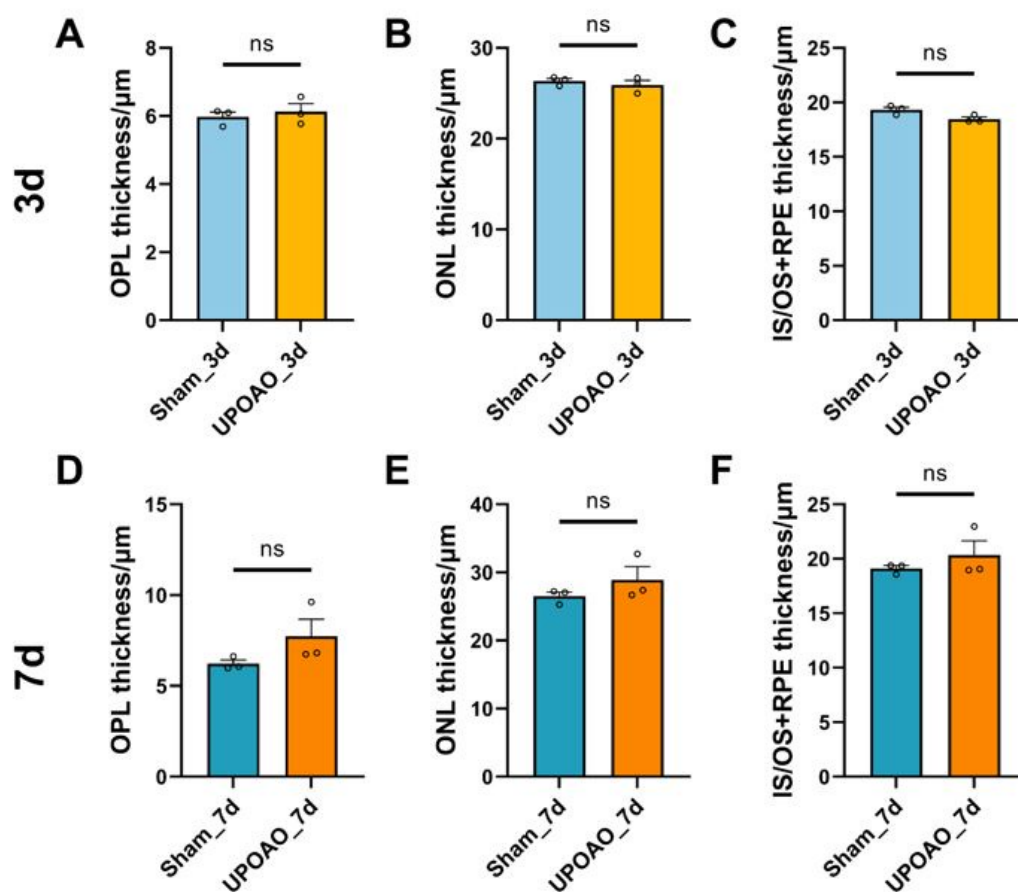
We aimed to convey that the amplitude of the a-waves, which represent the function of the photoreceptors, does not show significant variation, which is consistent with the thickness of the outer retinal layer observed in OCT and HE images. Our results indicated that at 7 days post-injury, the amplitude of the a-waves in ERG was statistically different only at stimulus light intensity of 0.3, 3.0 and 10.0 cd.s/m². In contrast, the b-wave amplitude was reduced by half compared to sham eyes at almost all stimulus light intensities. At the same time, the immunofluorescence staining results of photoreceptor cells showed no significant change at 7-days. Therefore, we consider the change in a-wave amplitudes were not significant compared to the significant decrease in b-wave amplitude. We have clarified this in the revised manuscript.

We also analyzed the thickness of the outer retinal layers in HE and found it to be consistent with OCT results, showing no significant changes (shown in below Author response image 2).

Thank you for your valuable feedback, which has helped improve the accuracy and clarity of our manuscript.

Author response image 2.

Thickness of OPL, ONL, IS/OS+RPE in HE staining. n=3; ns: no significance (p>0.05).



(10) Response to recommendation (10):

*Figure 5 and Figure S3: Quantification data from different sections of the same retina should be averaged to represent one single sample (one data point) for statistical analysis. * in images of Fig. 5E, F, I, J is not defined in the figure legend. It would be easier for readers to follow if the GCL, IPL, INL, and OPL were labeled in retinal sections.*

Thank you for your careful review of the manuscript and recommendation. We have reperformed the statistical analysis and updated the results in Figure 5 and Figure S3. In the UPOAO experimental eyes, no no significant change in the number of HCs (Calbindin) was observed during the 3-days reperfusion period, while a notable reduction was observed after 7 days (Figure 5). Additionally, we have added the definition of the asterisks (*) in the figure legend to clarify their significance. We have also labeled the retinal layers, including the GCL, IPL, INL, OPL, and ONL, in the images to make it easier for readers to follow and understand the data.

Thank you for helping us improve the clarity and accuracy of our manuscript.

(11) Response to recommendation (11):

Lines 407-409, the statement "which aligns with the a-waves observed in ERG (Figure 3D, E) and the changes seen in the outer retinal layers in OCT (Fig S2C, D)" is confusing. No changes were observed by OCT in Fig S2D.

Thank you for your review and we are sorry about the confusion. The overall trend of the amplitude of the a-wave in ERG at 7-days did not change significantly, which is consistent with the immunofluorescence staining results of the photoreceptor cells. Based on these observations, we consider that the change in the amplitude of the a-wave was not significant. As you pointed out in recommendation 9, since a-waves in ERG were changed in 7-days at the stimulus light intensity of 0.3, 3.0 and 10.0 cd.s/m², our description on the a-waves in 7-days was not accurate. We have clarified this point in the revised manuscript to ensure it accurately reflects the data presented.

(12) Response to recommendation (12):

In Figure S4, panel C shows lymphocyte-mediated immunity, and panel D shows leukocyte-mediated immunity. Please adjust the figure legend accordingly to reflect the figures.

Thank you for your careful review of the manuscript. We have modified the figure legend of Figure S4.

(13) Response to recommendation (13):

Lines 440-442 state "These results suggested early ischemic processions such as cell migration and potential collateral vessel formation." It is not clear why and how "potential collateral vessel formation" is suggested by Figure 6 and Figure S4. Please clarify this in the text.

Thank you for your careful review of the manuscript and we have deleted this sentence due to insufficient evidence. We have corrected this sentence: "These results suggested that in the early stage of retinal ischemic injury, leukocytes from the microvasculature may infiltrate retinal tissue. More experimental validation will be performed to confirm this hypothesis." (line 448). We will be more cautious in drawing conclusions in the future. Thank you for your reminder.

(14) Response to recommendation (14):

For the figure legend of Figure 6 "In each heatmap, upper box showed the top 10 up-regulated genes, and the below one showed the top 10 down-regulated genes." Is this correct? It appears that the upper box shows the top 10 down-regulated genes, and the lower box shows the top 10 up-regulated genes.

Thank you for your careful review of the manuscript and we have modified the figure legend of Figure 6. In the heatmaps, the upper box showed the top 10 down-regulated genes, and the below one showed the top 10 up-regulated genes (line 977).

(15) Response to recommendation (15):

For the figure legend of Figure 7, the statement 'Data points are from retinal sections of four animals' is incorrect, as these data were obtained from whole retinas instead of retinal sections. Please revise the legend to reflect this accurately. The scale bar was absent in the images of Figure 7. Asterisk in Figure 7H and 7I was not defined.

Thank you for your careful review of the manuscript and we have revised the errors. We have added the scale bar (Figure 7D). The white asterisks in Figure 7H and 7I indicate the activated microglial cells and we have added this definition in the legend of Figure 7 (line 981).

(16) Response to recommendation (16):

It would be better to switch the order of Figure S7 and Figure S8 to align with their descriptions in the text.

Thank you for your recommendation and we have switched the order of Figure S7 and Figure S8.

(17) Response to recommendation (17):

The gene names in Figure S8 should be written consistently with those listed in Table S1.

Thank you for your recommendation and we have corrected the gene names.

(18) Response to recommendation (18):

In Figure 9, it is not clear why amacrine cells were not included in the UPOAO model, as amacrine cells were also injured as shown in Figure 5I-L.

Thank you for your careful review of the manuscript and we have added amacrine cells in Figure 9.

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